SURF user manual

October 30, 2026 · Gergely Szarka

SURF (Spontaneous Undulation in Retina Finder) analyses retinal wave calcium-imaging videos, from the raw MP4 to wave statistics and group comparisons.

1. Getting started

What you need

- **Recordings:** wide-field calcium-imaging videos of whole-mount retinas (MP4, AVI, MOV or M4V), for example GCaMP recordings at 4×, 10× or 40×. Any frame rate works.
- **A computer:** Windows 10/11 or macOS (Apple Silicon or Intel). Cell detection needs about 3 GB of extra disk space.
- **For µm and µm/s:** an image or video of an object of known size taken through each objective, such as a stage micrometer or counting chamber (see section 2).

Workflow

Each video goes through the tabs from left to right. Every step saves itself, so you can stop at any point and continue later.

1. **Info:** enter the animal, age, condition, objective and eye, and check the video.
2. **Rotate:** mark the temporal pole, nasal pole and centre, so that the retina is dorsal up.
3. **Measure:** place a grid of squares over the retina and measure the brightness of every square in every frame.
4. **Events:** detect calcium events in every square.
5. **Waves:** find the wavefronts, track them and calculate frequency, speed, duration and direction.
6. **Cells:** at high enough magnification, find single cells and test which ones take part in the waves.
7. **Compare:** pool the results of many videos and compare ages or conditions.

Installing

Computer	How
Windows	Unzip SURF.zip and double-click **install_windows.bat**. Start SURF from the desktop or Start-menu shortcut.
macOS	Unzip SURF.zip and run **install_mac.command**. If macOS blocks it, open System Settings → Privacy & Security and click **Open Anyway**. SURF then appears in Applications.
Cell detection (optional)	In the Cells tab, click **Install Suite2p + Cellpose**. On Windows SURF closes, installs in its own window and restarts.

To update, unzip the new SURF.zip and run the installer again. Settings, presets and results are kept. **Help → About** shows the version you are running.

Opening videos

Click **Open folder…** (top left) and choose the folder with your recordings. The video list shows a thumbnail of every video, and under it:

- **animal · age · condition · objective**, once they are entered on the Info tab;
- tags that show the progress: **[calib]** rotation marked, **[measured]** grid measured, **[cells]** cells found.

Click a video to select it; every tab then works on that video. The panel under the list shows its size, frame rate and length. **Show video in folder** opens the folder in Explorer or Finder.

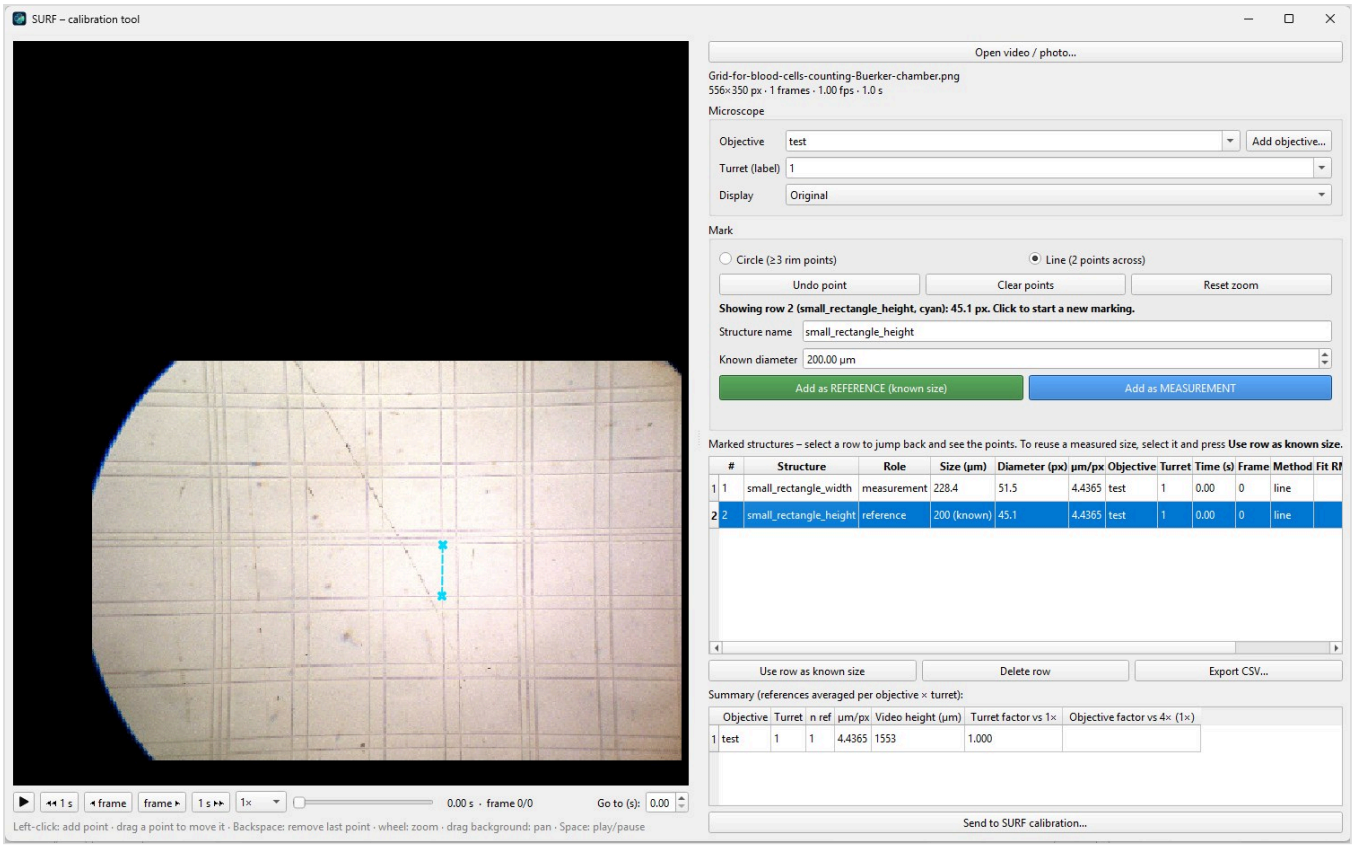
Files that SURF writes

File	Contents
`<video>_wavegui.json`	all settings and recording information for the video (it is read again when you reopen it)
`<video>_grid` or `<video>_rotated_grid` folder	the measurement, `…_wave_results.xlsx` workbook, figures (PNG, SVG, EPS, PDF) and cell results
`<video>_dFoF.mp4`, `<video>_rotated.mp4`	optional ΔF/F and rotated videos

The original video is never changed.

2. Spatial calibration (Tools → Calibration tool)

The calibration tool converts pixels to µm for each objective and turret setting. You only need to do this once per microscope setup; SURF then shows every distance and speed in µm.



Calibration tool with a counting-chamber grid: a 200 µm line marked as reference

1. **Open video / photo…**: load an image or video of something of known size, taken through the same objective. Examples are a counting-chamber grid, a stage micrometer, or a ring of known diameter.
2. Choose the **Objective** and **Turret** (magnification changer) it was taken with. **Add objective…** adds a new objective name.
3. Choose how to mark it: **Line** (click the two ends) or **Circle** (click 3 or more points on the rim; a circle is fitted to them).
    - Wheel zooms, dragging the background pans and dragging a point moves it.
    - **Backspace** removes the last point; **Undo point** and **Clear points** do the same with buttons.
4. Type the **Structure name** and its **Known diameter** in µm.
5. Click **Add as REFERENCE (known size)**. The table shows the length in pixels and the resulting µm/px.
6. Optionally mark other structures with **Add as MEASUREMENT**, to check the calibration. The tool calculates their size from the reference; it should match their real size.
7. Check the **Summary**: the averaged µm/px and video height per objective and turret, and the turret factors.
8. Click **Send to SURF calibration…** to store the values. SURF uses them for every video with that objective and turret.

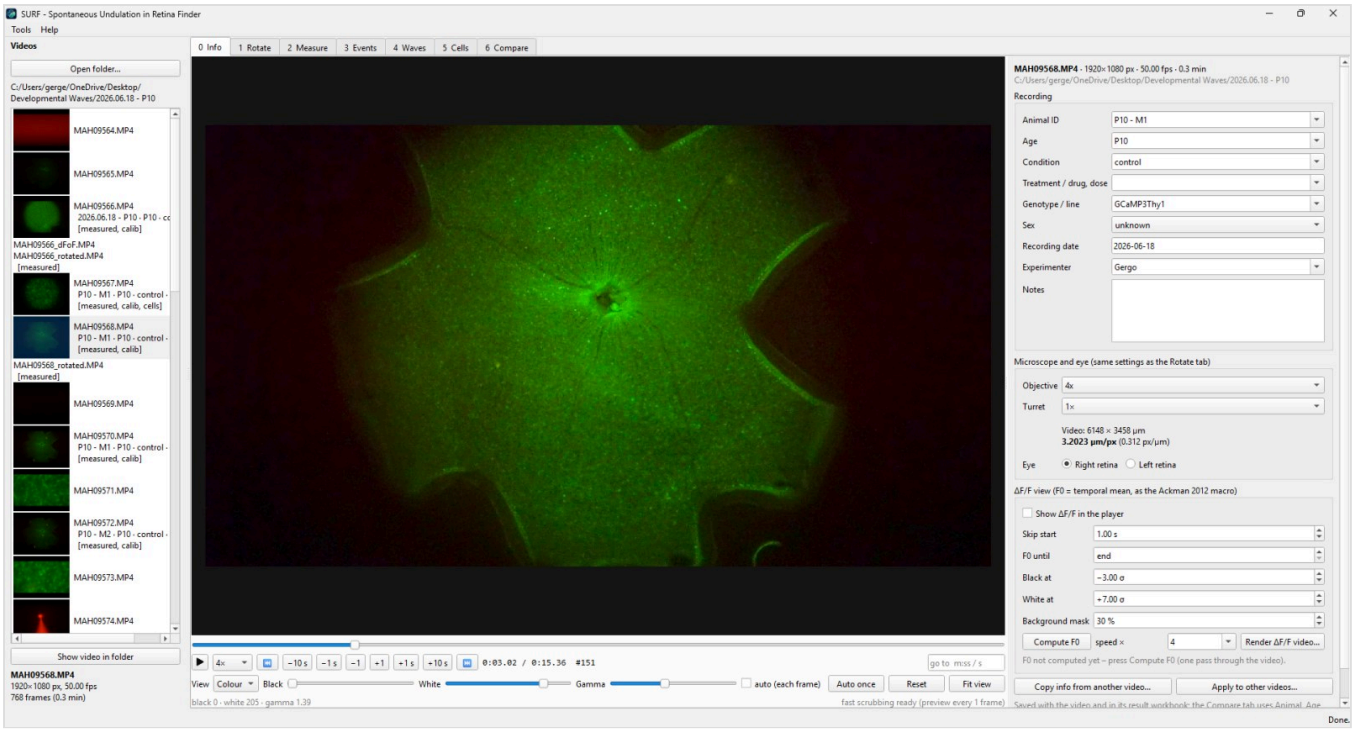
Tips:

- Select a row in the table to see its points again (cyan).
- **Use row as known size** turns a measured structure into a reference.

- **Export CSV…** saves the table.
- The marked points are saved next to the image (`…_calib.json`).

3. Tab 0 – Info

The Info tab holds what the recording is: who, what age, which condition and which objective. The Compare tab later groups the videos by these fields, so fill them in for every video.



Info tab: video player on the left, recording details on the right

Recording details

- **Animal ID**, **Age**, **Condition**, **Treatment / drug, dose**, **Genotype / line**, **Sex**, **Recording date**, **Experimenter**, **Notes**. The menus offer the values used before, so names stay consistent.
- Use the same Animal ID for every video of one animal: statistics in Compare are done per animal.
- **Copy info from another video…**: click it, then click the source video in the list.
- **Apply to other videos…**: writes these fields to several videos at once.

Microscope and eye

Objective and **Turret** set the scale. The panel shows the frame size in µm and the µm/px value, for example 6148 × 3458 µm and 3.20 µm/px for 4× at turret 1×. **Eye** (right or left retina) is needed for the anatomical directions. These are the same settings as on the Rotate tab; changing one changes the other.

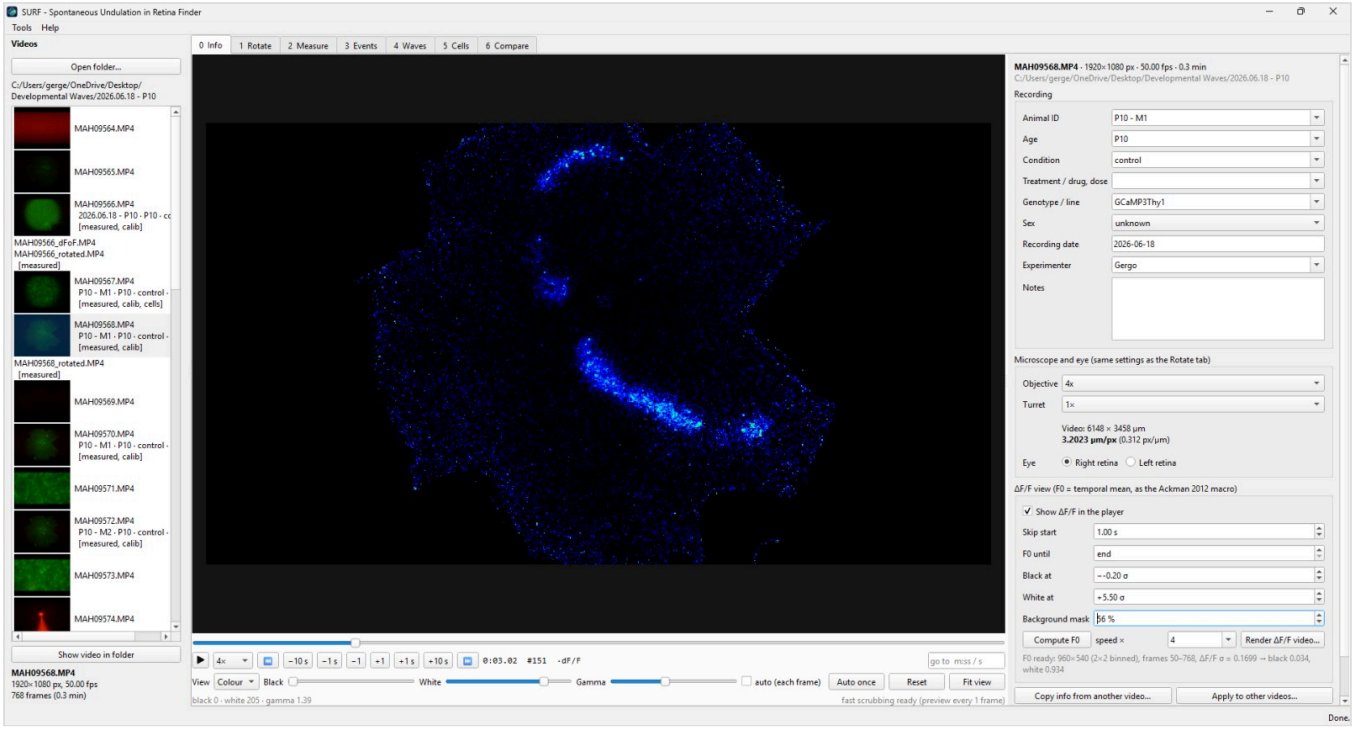
Video player

Control	What it does
▶ and speed menu	play at 0.25× to 64×
−10 s … +10 s, −1 / +1	jump in time or by single frames
slider, *go to m:ss / s*	scrub, or type a time to jump to
View	Colour, Grey or one channel (Green, Red, Blue)
Black, White, Gamma	brightness and contrast
auto (each frame), Auto once, Reset	automatic contrast continuously, once, or back to default
Fit view	zoom back to the whole frame (wheel zooms, drag pans)

The status line says when **fast scrubbing** is ready. SURF builds a small preview of the whole video in the background, after which the slider moves instantly.

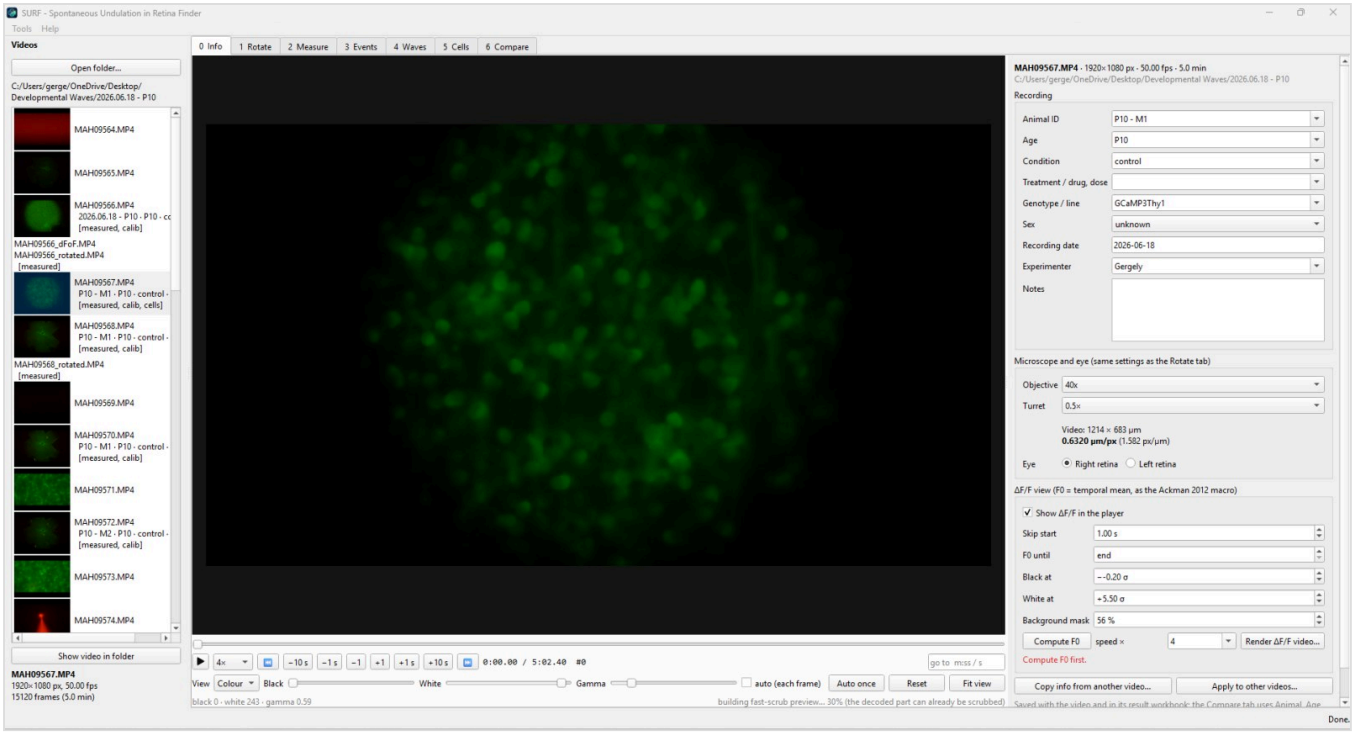
ΔF/F view

ΔF/F shows where activity happens, which raw fluorescence hides. F0 is each pixel's mean over time, as in the Ackman et al. (2012) macro.



ΔF/F view of a 4× recording: a wave in blue on black

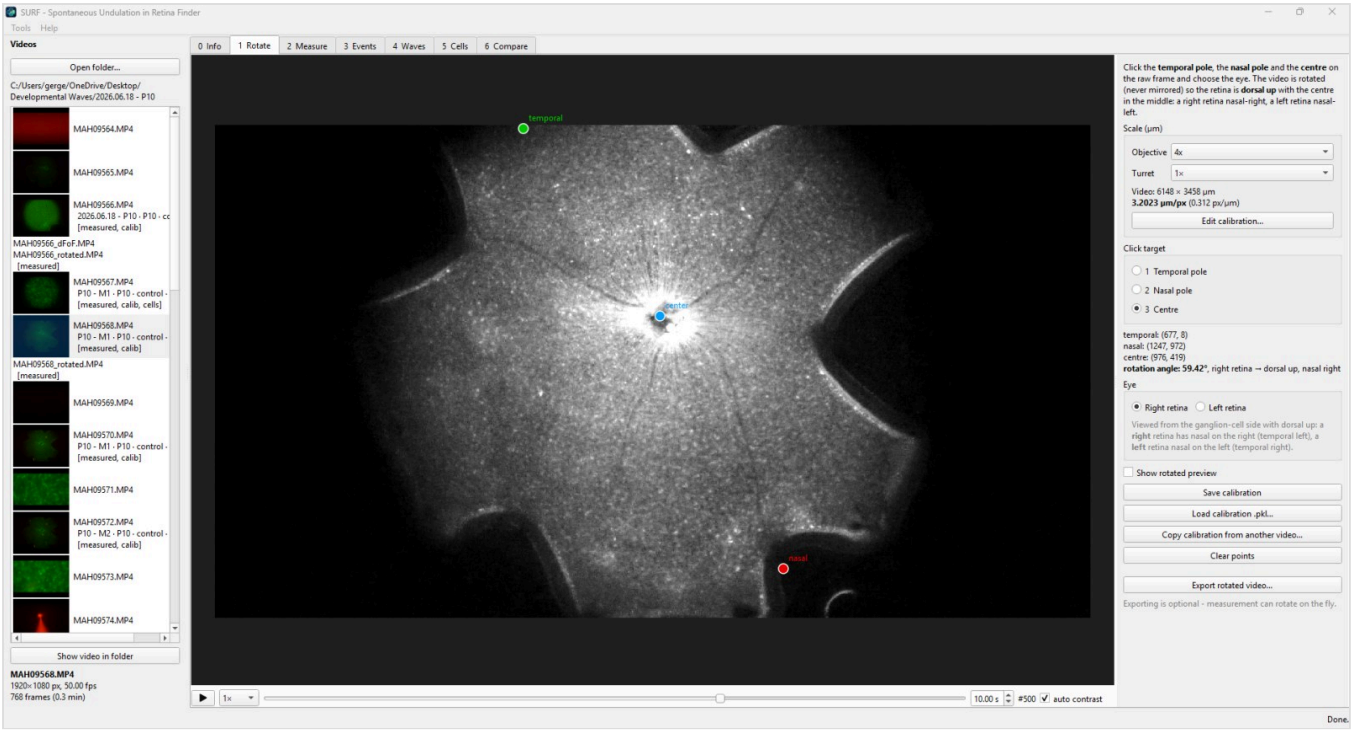
1. Set **Skip start** (seconds left out at the beginning) and **F0 until** (*end* = the whole video).
2. Click **Compute F0**. This is one pass through the video, and the result is saved, so it is needed only once.
3. Tick **Show ΔF/F in the player**.
4. Adjust the display:
    - **Black at** and **White at** set the colour range in units of the ΔF/F standard deviation (σ). Raise *Black at* to hide noise.
    - **Background mask** (%) blacks out the dimmest pixels, outside the retina.
5. **Render ΔF/F video…** writes an MP4 at the chosen **speed**.



Info tab of a 40× recording (40×, turret 0.5×: 1214 × 683 µm, 0.632 µm/px)

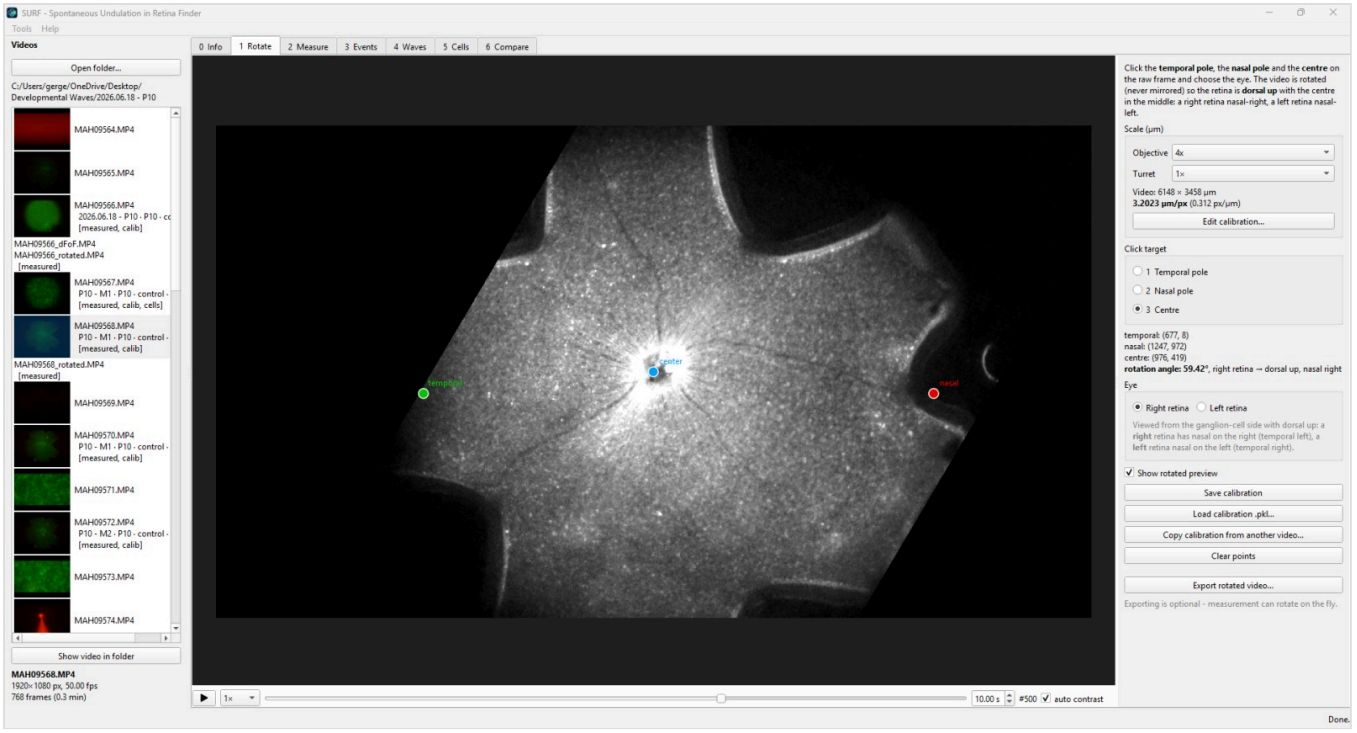
4. Tab 1 – Rotate

Rotation turns every recording to the same view, dorsal up, so wave directions can be compared between retinas. Viewed from the ganglion-cell side with dorsal up, a **right retina has nasal on the right** and a **left retina has nasal on the left**. Retinas are only rotated, never mirrored.



Rotate tab: temporal (green), nasal (red) and centre (blue) marked on the raw frame

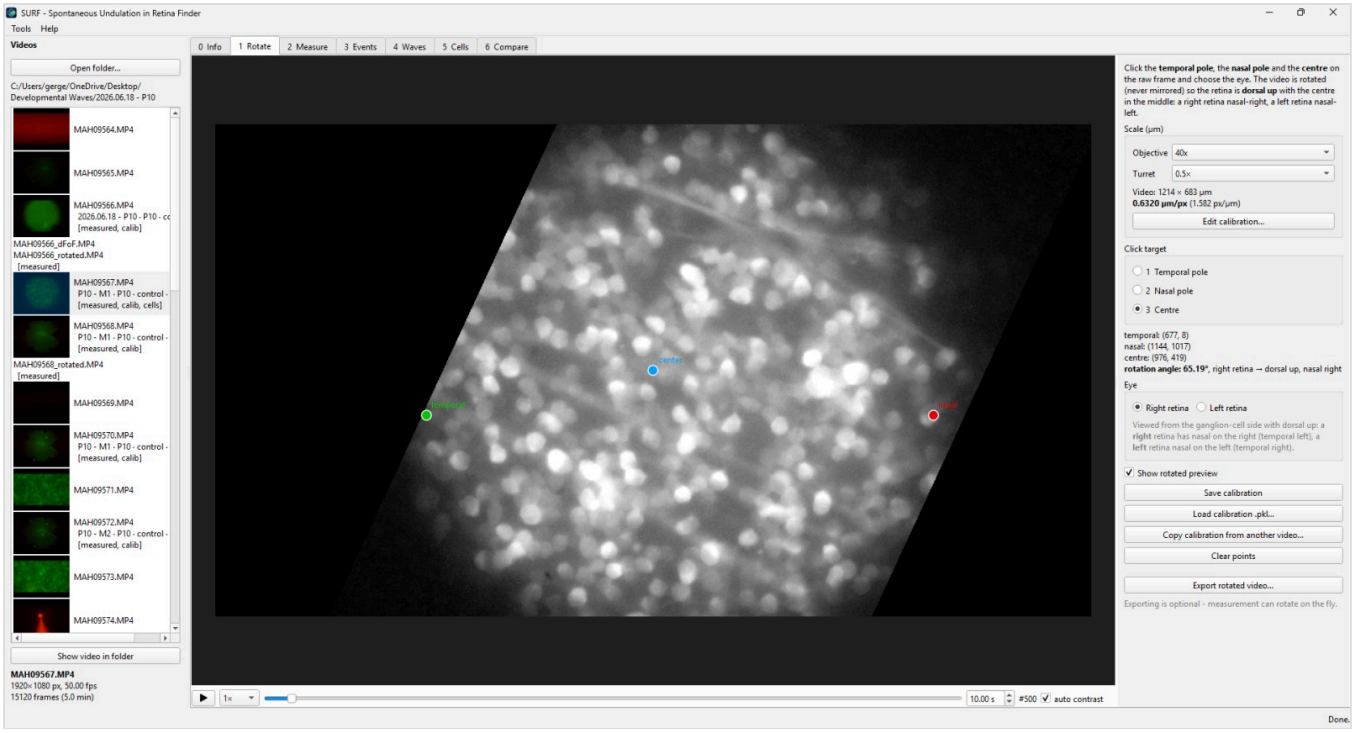
1. Choose the **Eye** (Right or Left retina).
2. Under **Click target**, pick **1 Temporal pole** and click it on the frame, then **2 Nasal pole**, then **3 Centre** (usually the optic disc).
    - Points can be dragged afterwards.
    - The panel shows their coordinates and the **rotation angle**, for example 59.42°, right retina → dorsal up, nasal right.
3. Tick **Show rotated preview** to check the result: temporal and nasal should be level, on the correct sides, with the centre in the middle.
4. Click **Save calibration**. The list shows **[calib]** for this video.



Rotated preview: temporal left, nasal right, dorsal up

Button	Use
Copy calibration from another video…	click it, then the source video. Use this for recordings of the same retina in the same position.
Load calibration .pkl…	reads the rotation file of the original notebook
Clear points	start again
Export rotated video…	optional. Measurement rotates on the fly, so this only matters if you want the rotated file.
Edit calibration…	opens the µm/px settings of the Scale panel

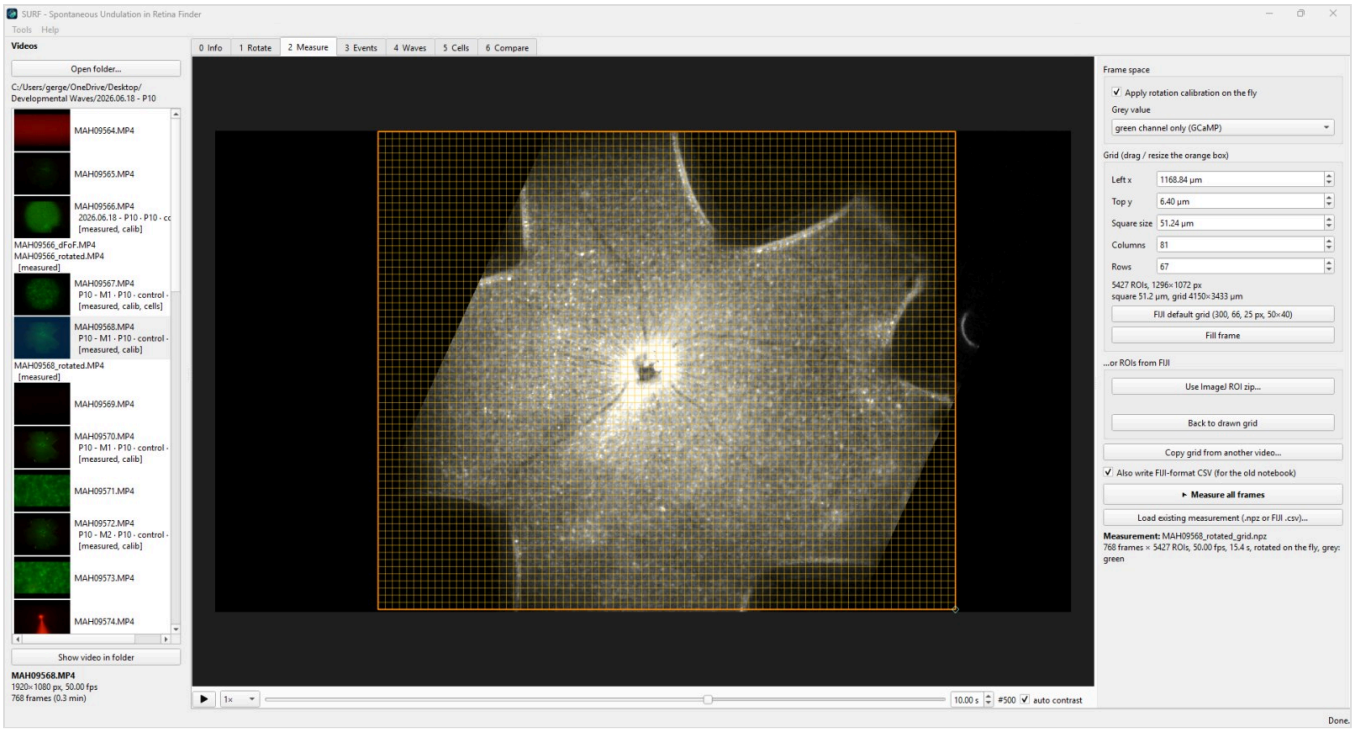
At 40× the poles are usually outside the field. Copy the calibration from the 4× recording of the same retina, if the retina and camera were not moved between the recordings. Then check the rotated preview.



40× recording with the orientation copied from the 4× video of the same retina

5. Tab 2 – Measure

Measure covers the retina with a grid of square ROIs and reads the mean brightness of every square in every frame. This replaces the FIJI Multi Measure step and takes about a minute for a 5-minute video.

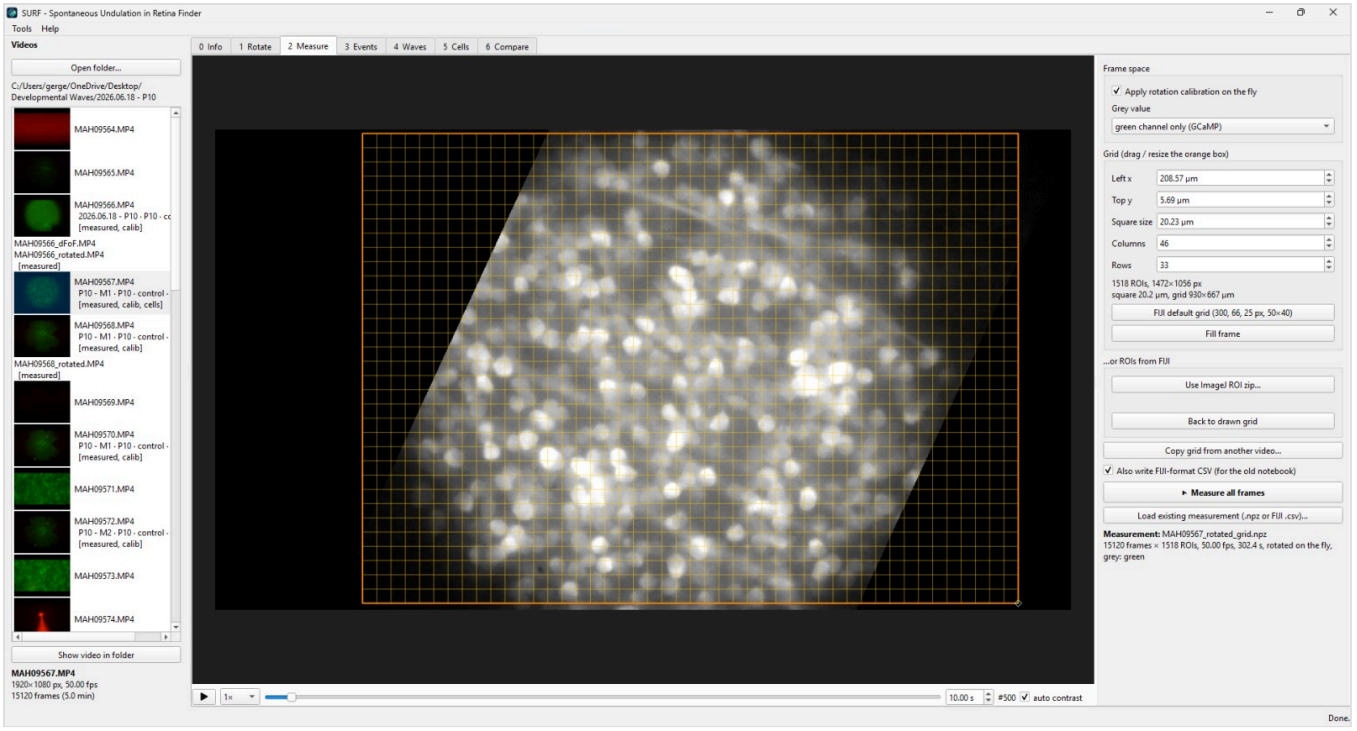


Measure tab, 4×: 81 × 67 squares of 51.2 µm over the rotated retina

1. **Frame space:** leave **Apply rotation calibration on the fly** ticked, so the grid is measured on the rotated retina.
2. **Grey value:** choose **green channel only (GCaMP)** for GCaMP recordings. The other options are the mean of the colour channels, or weighted luminance.
3. **Grid:** drag the orange box to move it and drag its corner to resize it, or type **Left x**, **Top y**, **Square size**, **Columns** and **Rows** (in µm once the scale is set).
    - The line below shows the number of ROIs and the grid size.
    - **Fill frame** covers the whole frame.
    - **FIJI default grid** restores the notebook grid (300, 66, 25 px, 50 × 40).
4. Click **Measure all frames**. The status line shows the result, for example *768 frames × 5427 ROIs, 50 fps, rotated on the fly, grey: green*. The video list shows **[measured]**.

Option	Use
Use ImageJ ROI zip…	measure ROIs drawn in FIJI instead of the grid; **Back to drawn grid** returns to the grid
Copy grid from another video…	the same grid on several recordings of one retina
Also write FIJI-format CSV	an extra file in the format of the old notebook
Load existing measurement…	reopen an earlier `.npz` measurement or a FIJI `.csv`

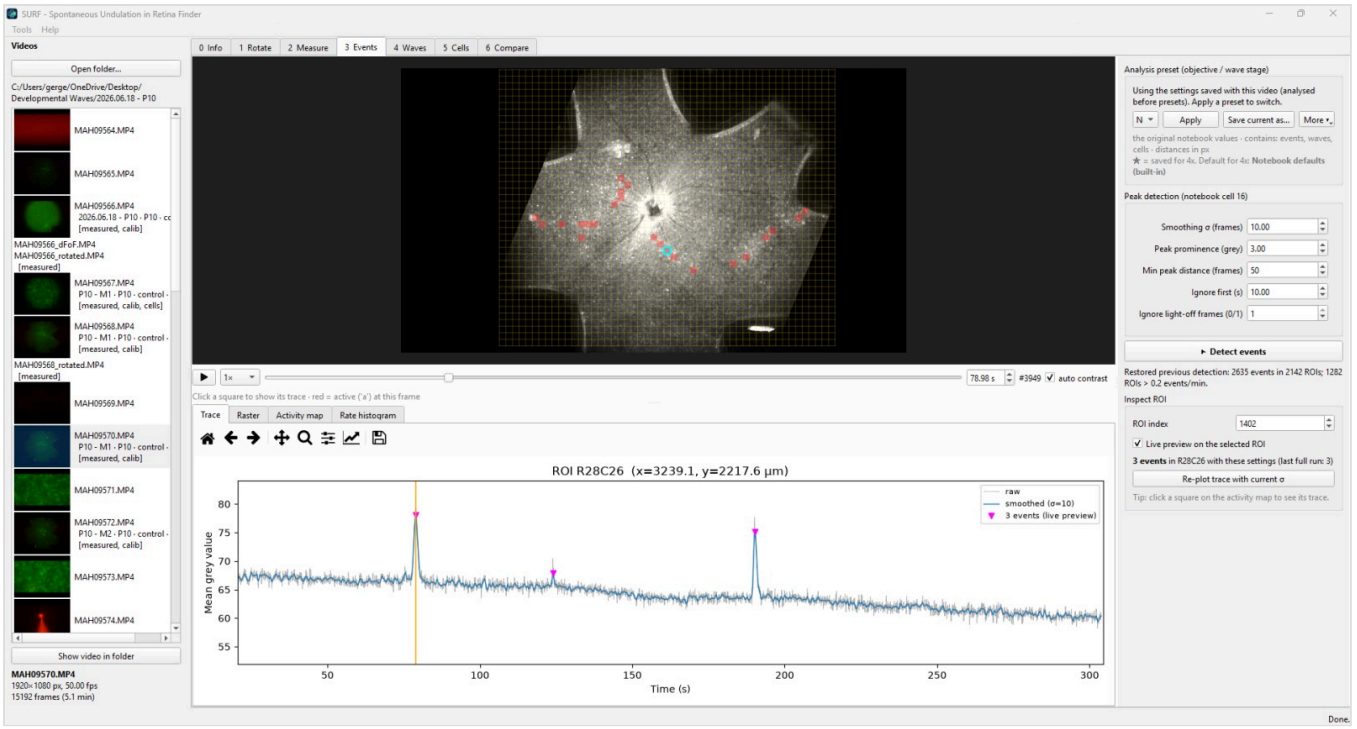
The **square size** sets the spatial resolution. Typical sizes are about 50 µm at 4× and about 20 µm at 40×. Analysis presets set the size automatically for each objective (see the Events tab).



Measure tab, 40×: 46 × 33 squares of 20.2 µm

6. Tab 3 – Events

Event detection finds the calcium transients in every square. The trace is smoothed, and peaks that rise far enough above their surroundings count as events.



Events tab, 4×: video with active squares (red), trace of the selected square below, settings on the right

Analysis presets

A preset is a saved set of settings for one objective and wave stage, for example *4× – stage II waves* or *40× – Stage II*. It holds the event, wave and cell settings and the grid size. Event detection runs only after a preset has been applied. The default preset for the video's objective is pre-selected.

- Choose a preset and click **Apply**. If you change settings afterwards, the panel says *modified*.
- **Save current as…** stores the current settings as a new preset.
- **More ▾**: make a preset the default for an objective, delete it, or export and import presets to share them with colleagues.
- *Notebook defaults (built-in)* are the values of the original notebook.

Peak detection

Setting	Meaning	Recommended start, 4×	Recommended start, 40×
Smoothing σ (frames)	Gaussian smoothing of the trace before peak finding	10	8
Peak prominence (grey)	how far a peak must rise above its surroundings	3.0	1.9
Min peak distance (frames)	the closest two events in one square may be	50	50
Ignore first (s)	start of the recording left out (onset artefact)	10	6
Ignore light-off frames (0/1)	1 = frames with the light off are bridged and no event is counted next to them	1	1

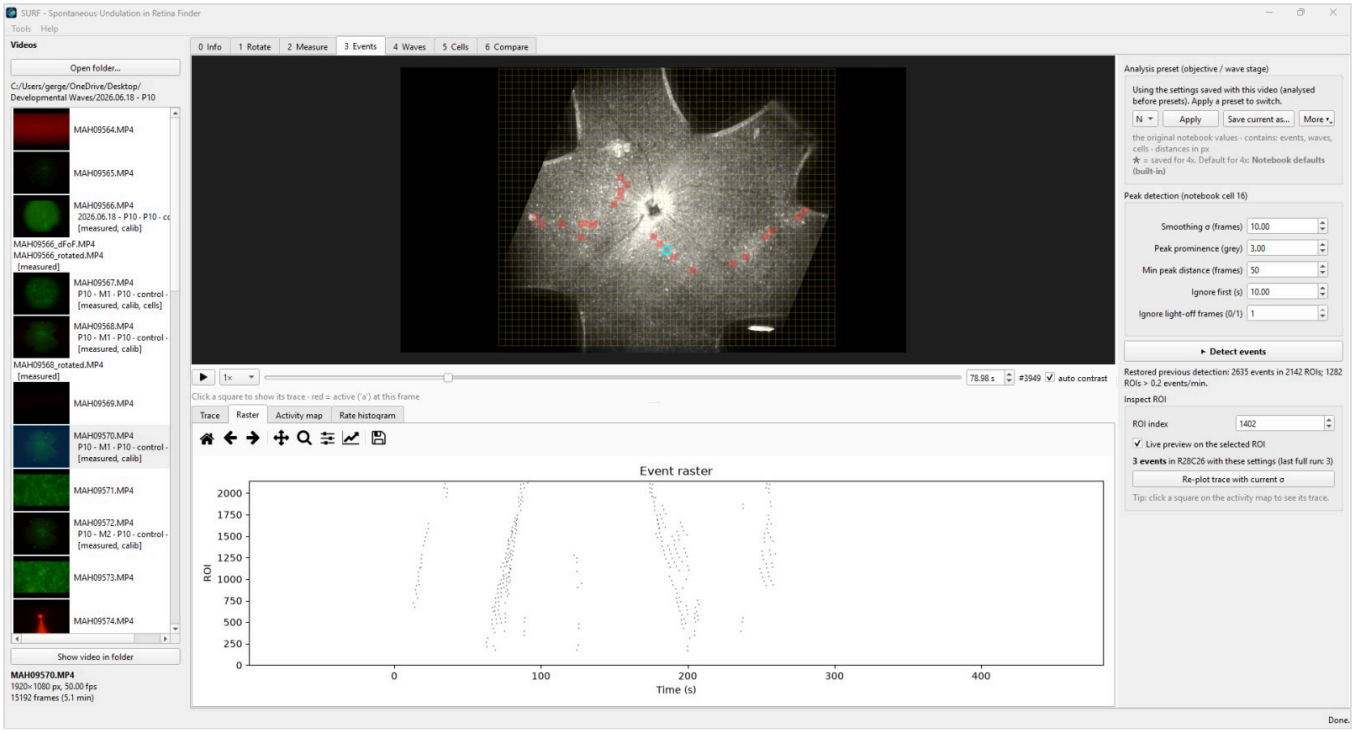
The recommended values are a starting point for GCaMP recordings at 50 fps. Check them with the live preview on a few squares, adjust them for your recordings, and save them as a preset.

Click **Detect events**. The panel reports the result, for example *2635 events in 2142 ROIs; 1282 ROIs > 0.2 events/min*.

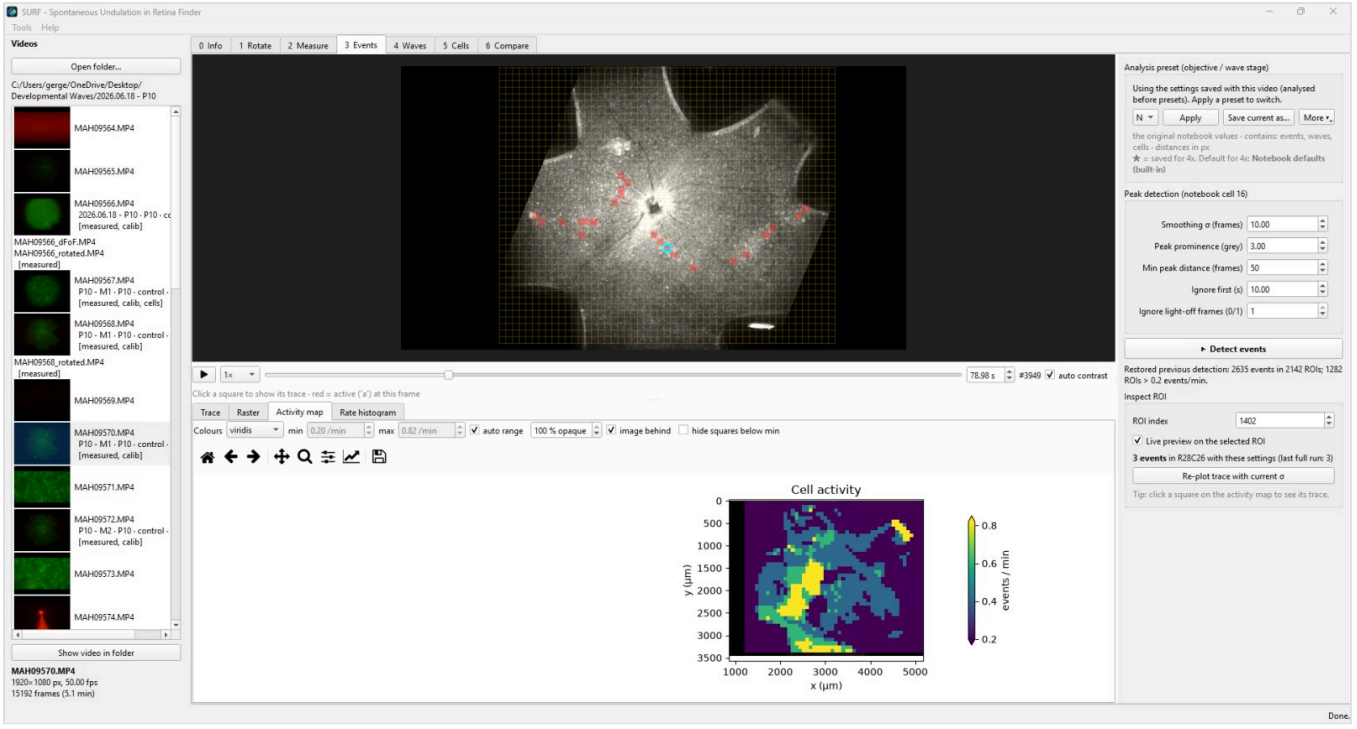
Checking the detection

- **Click a square** on the video (or set its **ROI index**) to see its trace. Red squares are active at the current frame.
- **Live preview on the selected ROI** redraws the trace and its events (magenta ▼) while you change a setting, so you can tune the settings on a few squares before running everything.
- The orange line in the trace is the current video frame.

Plot	Shows
Trace	raw (grey) and smoothed (blue) trace of one square, with its events
Raster	events of all squares over time. Waves appear as slanted lines, because squares are activated one after the other.
Activity map	events per minute in every square. **Colours**, **min / max** or **auto range**, opacity, image behind and *hide squares below min* adjust it. Click a square to open its trace.
Rate histogram	distribution of event rates



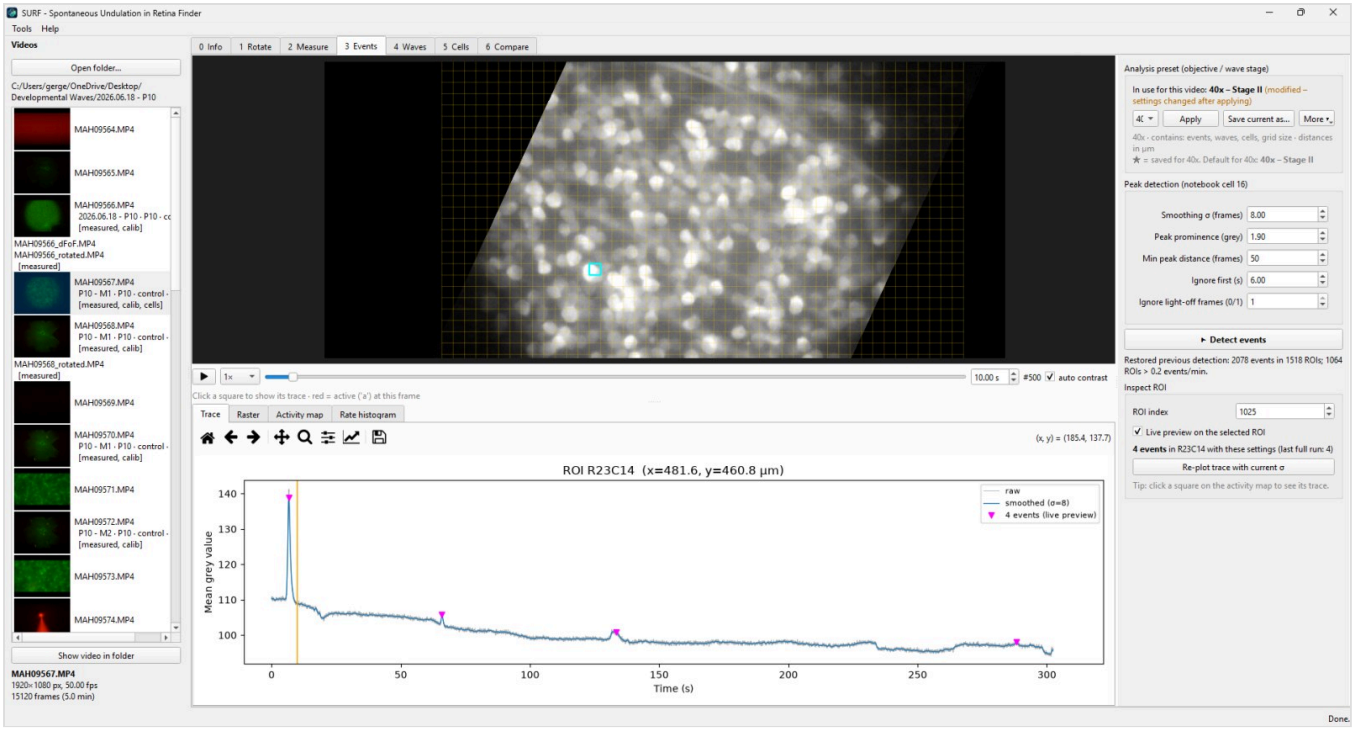
Raster, 4×: each slanted band is a wave crossing the retina



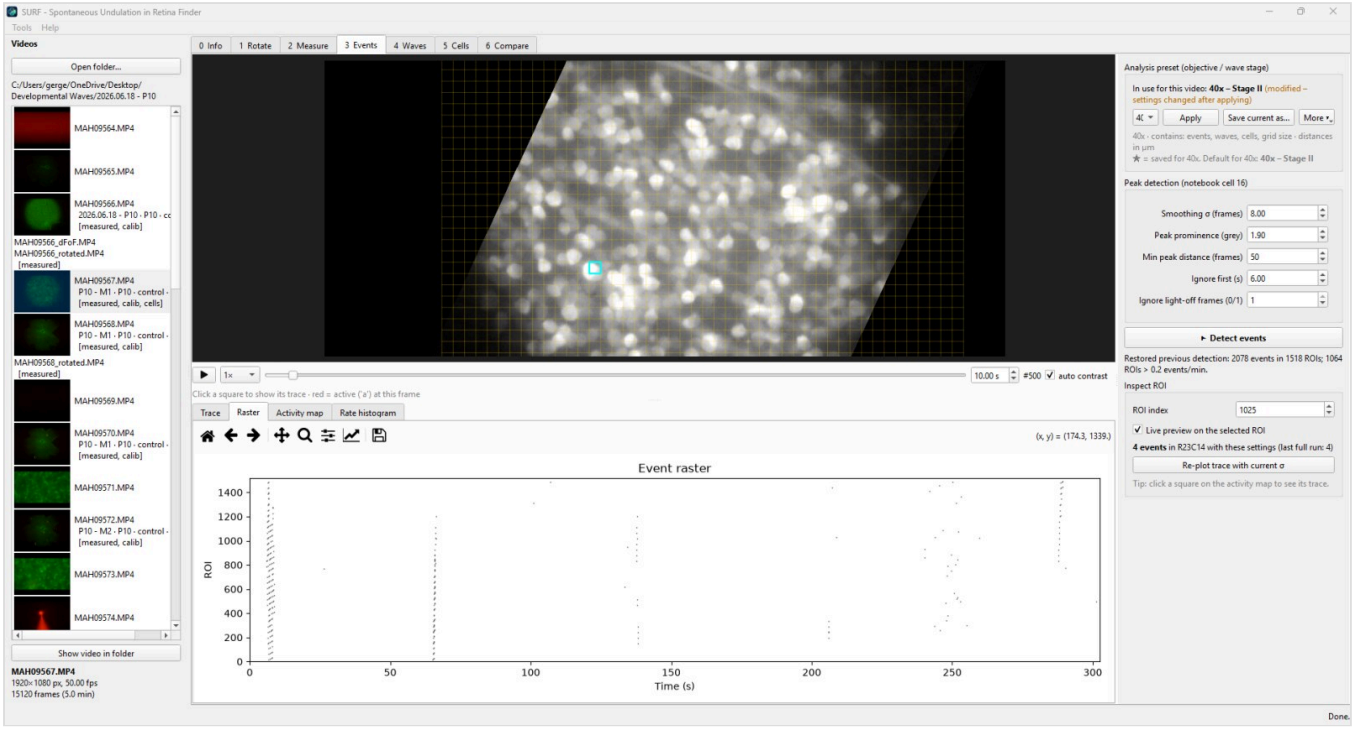
Activity map, 4×: event rate per square over the retina

All plots zoom the same way:

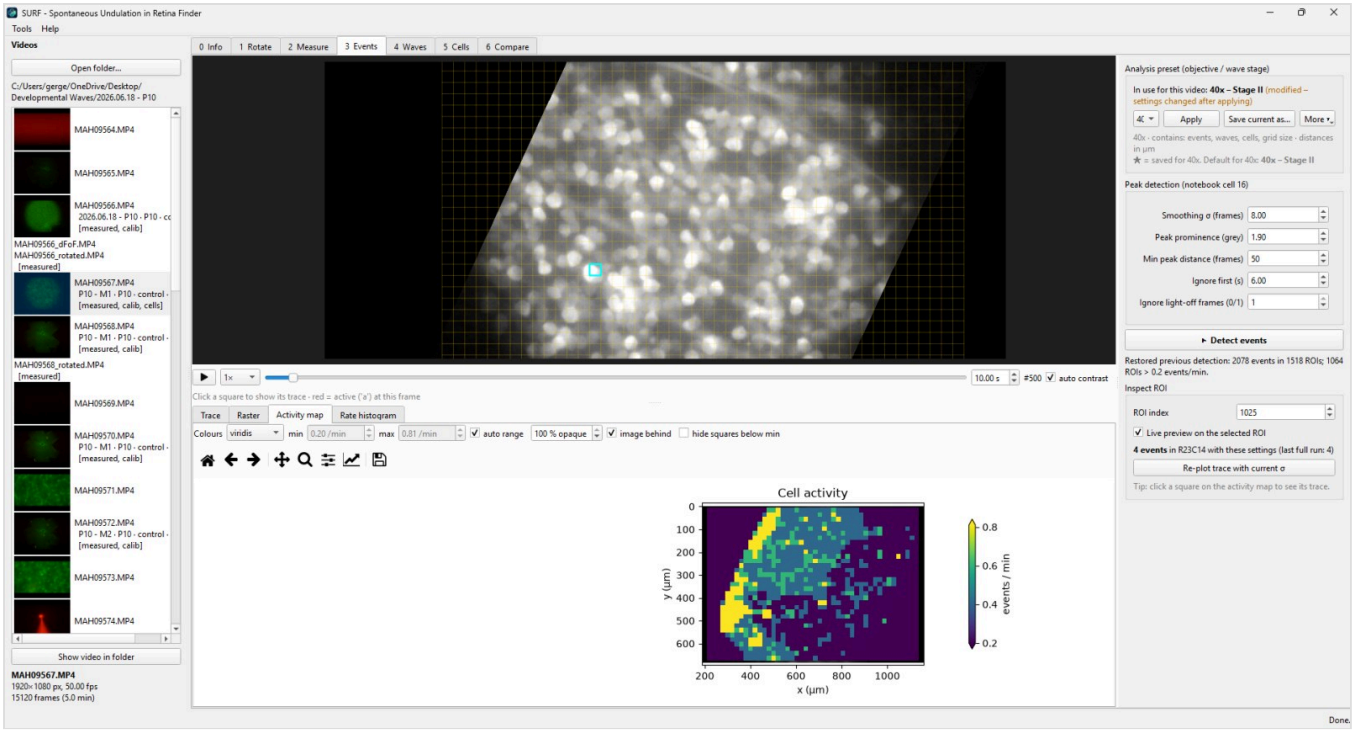
- drag along an axis to pan it, and use the wheel over an axis to zoom it;
- right-drag stretches an axis;
- double-click resets the view.



Events tab, 40×: trace of one square with four events



Raster, 40×: waves are near-vertical lines, because they cross the small field quickly



Activity map, 40×

7. Tab 4 – Waves

The wave analysis turns the active squares into moving wavefronts:

1. In every frame, the active squares are grouped into wavefront **centres** (k-means clustering).
2. Centres are linked from frame to frame into **tracks**.
3. From the tracks SURF calculates frequency, duration, speed and direction.

Click **Run wave analysis**, then **Save results (xlsx + figures)**.

Retinal orientation

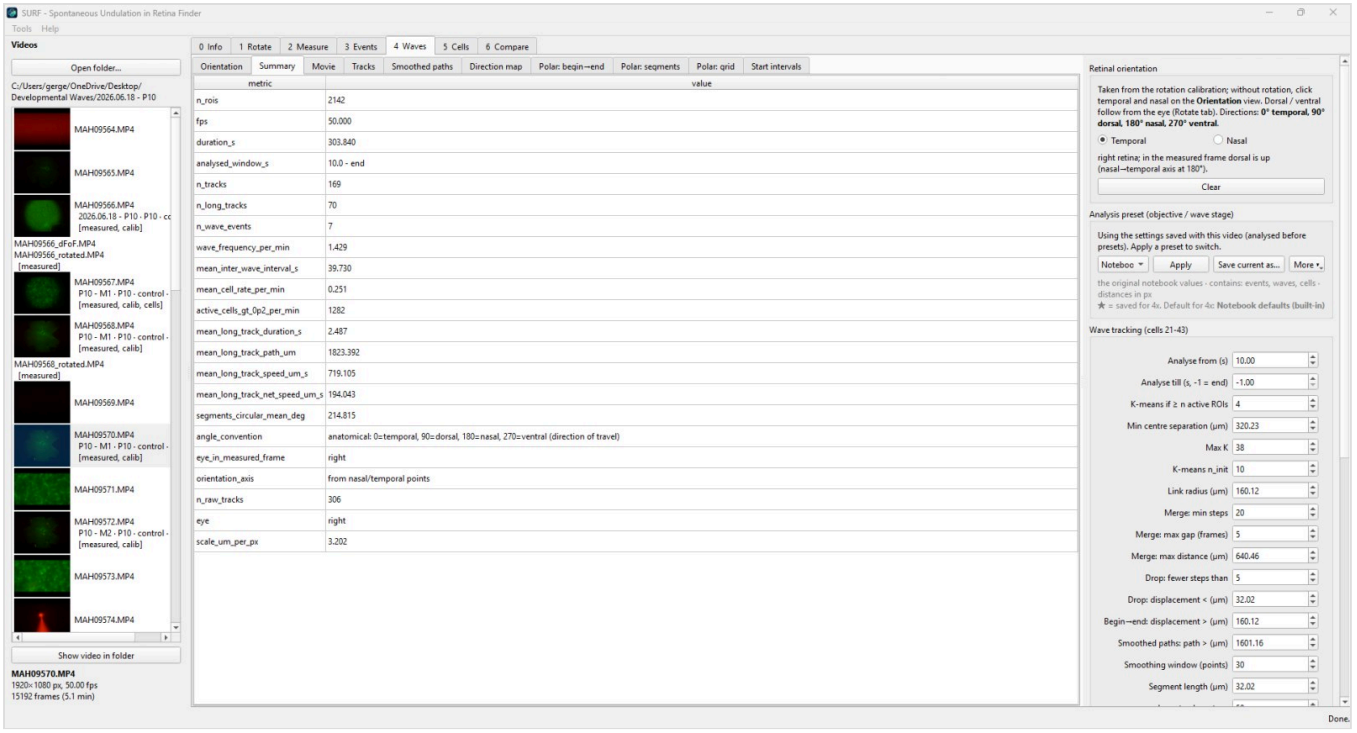
The orientation comes from the Rotate tab. The panel confirms it, for example *right retina; in the measured frame dorsal is up*. Directions are anatomical: **0° temporal, 90° dorsal, 180° nasal, 270° ventral** (direction of travel). Without a rotation calibration, click temporal and nasal on the **Orientation** view.

Wave tracking settings

Distances are in µm. The preset fills them in; change them here when needed.

Setting	What it controls
Analyse from / till (s)	time window analysed (−1 = to the end)
K-means if ≥ n active ROIs	below this, the centre is simply the average position of the active squares
Min centre separation	separate centres must be at least this far apart. **Raise it if one wave is split into several centres; lower it if separate waves are merged.**
Max K, K-means n_init	at most this many centres per frame; clustering restarts
Link radius	how far a centre may move between frames and stay the same track
Merge: min steps / max gap (frames) / max distance	joins track pieces that belong together
Drop: fewer steps than / displacement <	removes very short or non-moving tracks
Begin→end: displacement >	tracks used for the begin-to-end direction plot
Smoothed paths: path >, Smoothing window, Segment length	tracks used for the smoothed paths and segment directions
Long track: > steps	tracks counted as waves for speed and duration
New wave if start gap > (s)	track starts further apart than this count as separate wave events
Grid map binning (ROIs)	coarseness of the direction map
Worker processes	CPU cores used; more = faster

Results

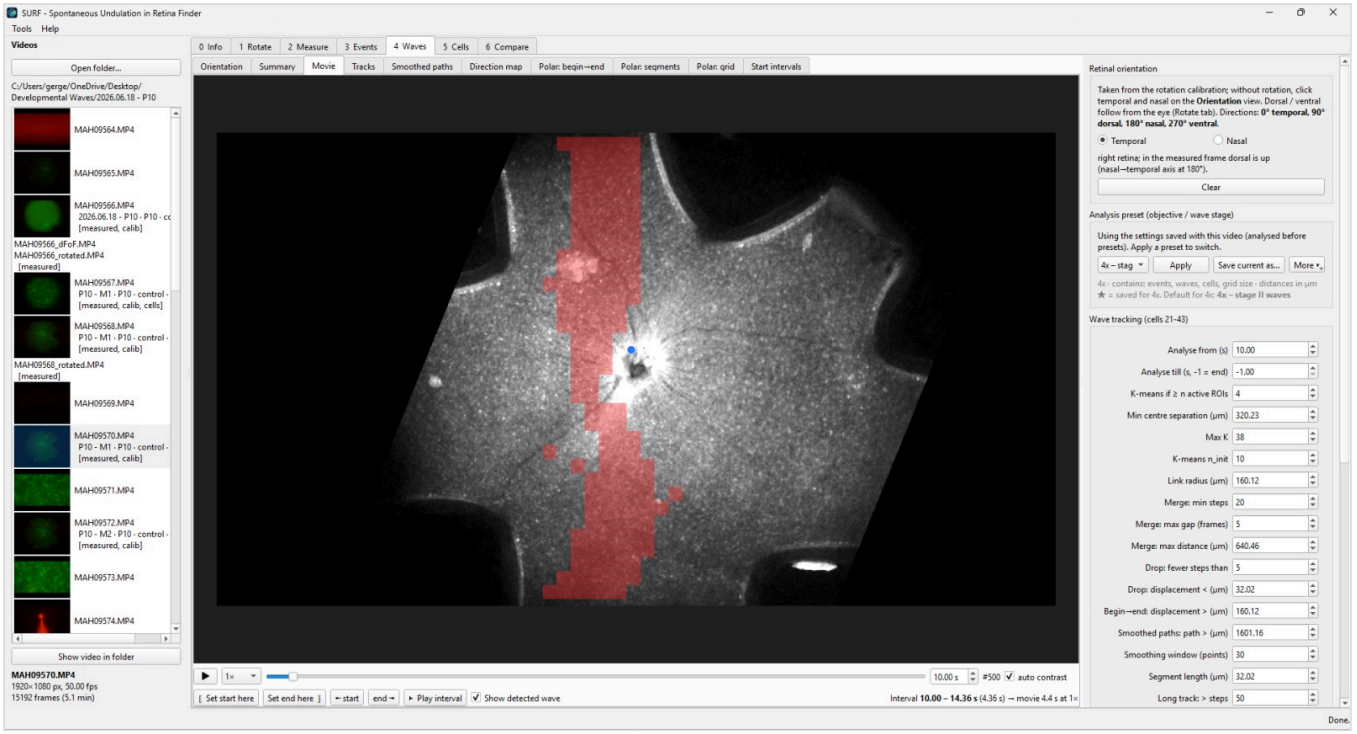


Summary: one row per result (number of tracks and waves, frequency, interval, duration, path length, speed, direction)

The main numbers in **Summary** are:

- **n_wave_events** and **wave_frequency_per_min**: number of waves and waves per minute;
- **mean_inter_wave_interval_s**: time between wave starts;
- **mean_long_track_duration_s**, **_path_um**, **_speed_um_s**, **_net_speed_um_s**: wave duration, path length, speed along the path and start-to-end speed;
- **segments_circular_mean_deg**: mean direction;
- **mean_cell_rate_per_min** and **active_cells_gt_0p2_per_min**: event rate per square, and the number of active squares.

Movie



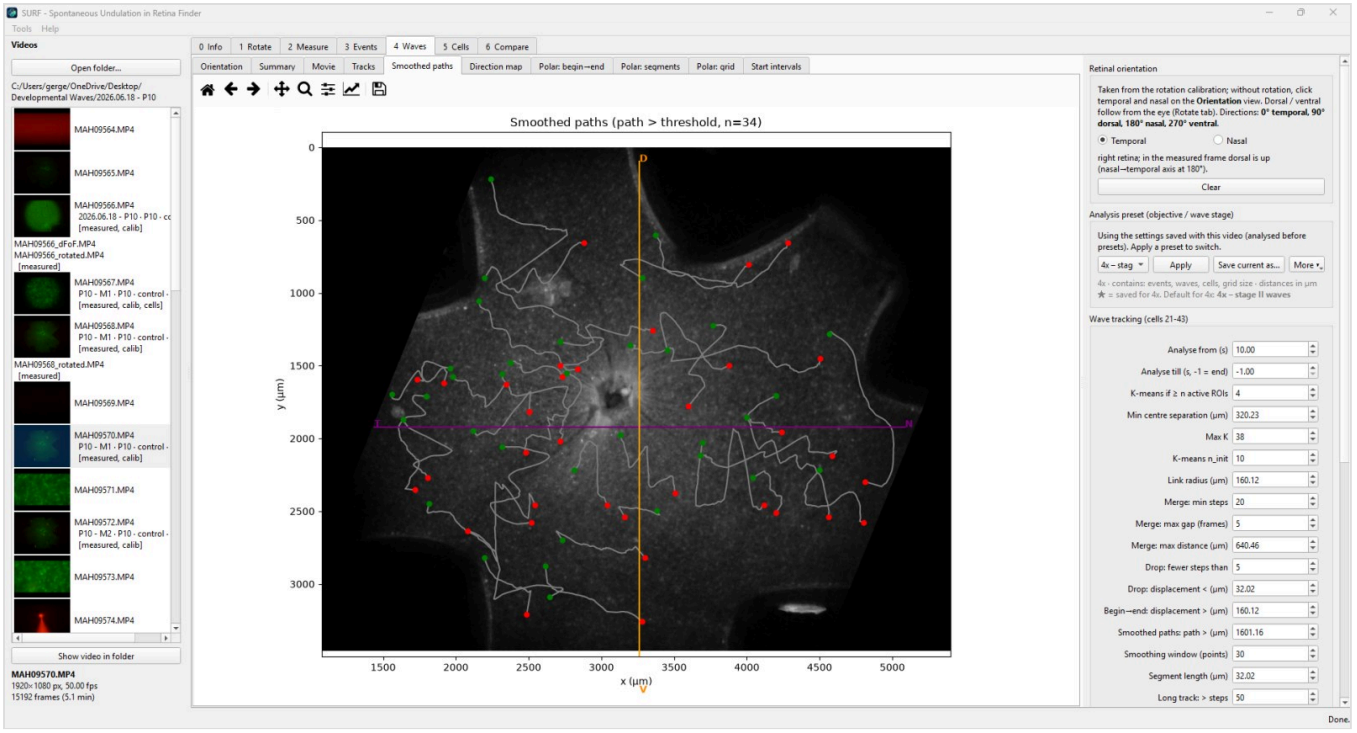
Movie: original video with the detected wave (red squares) and its centre (blue)

The **Movie** view plays the video with the detected wave on top, so you can check the analysis by eye.

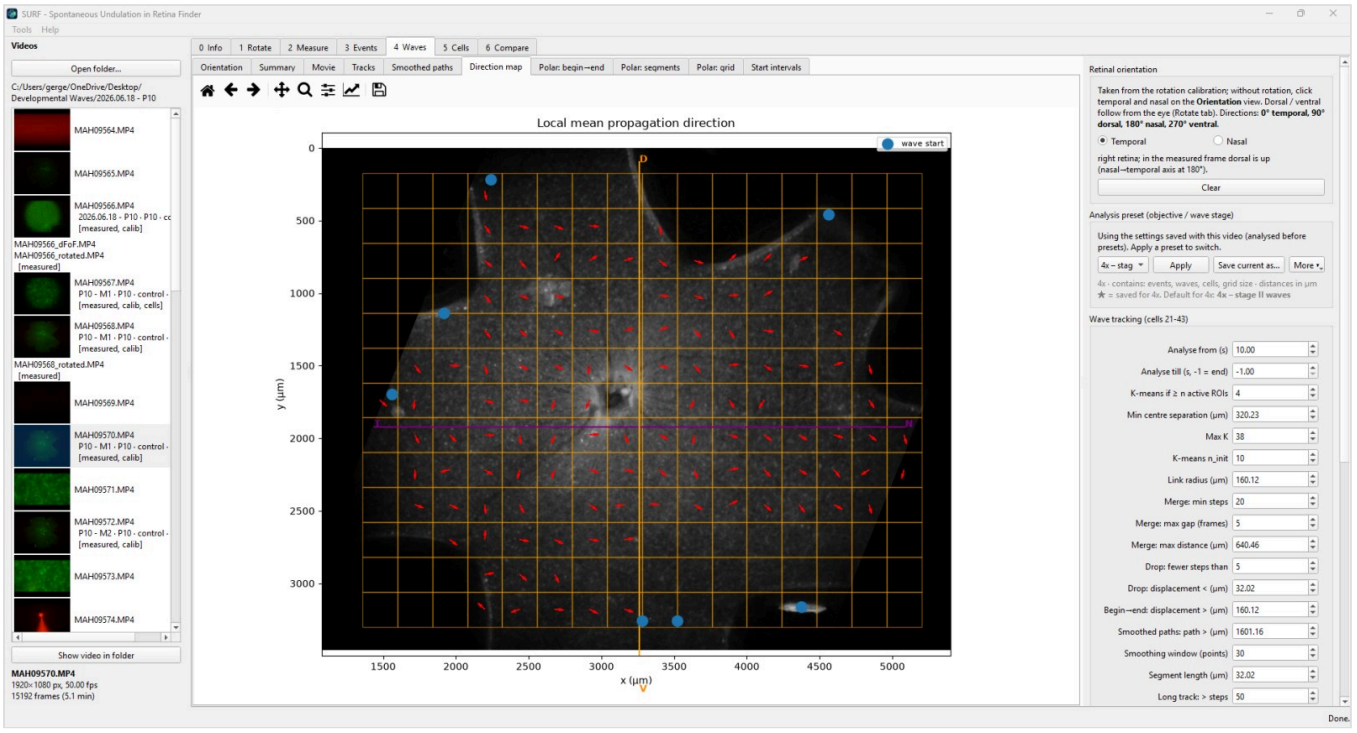
- **Show detected wave** draws the active squares (red) and the centres (blue).
- **Set start here** / **Set end here** choose an interval; **Play interval** plays it.
- The Movie panel on the right renders it as an MP4: *full* shows raster states, centres and tracks; *filtered* shows active squares only. You set the time label and the speed there.

Plots

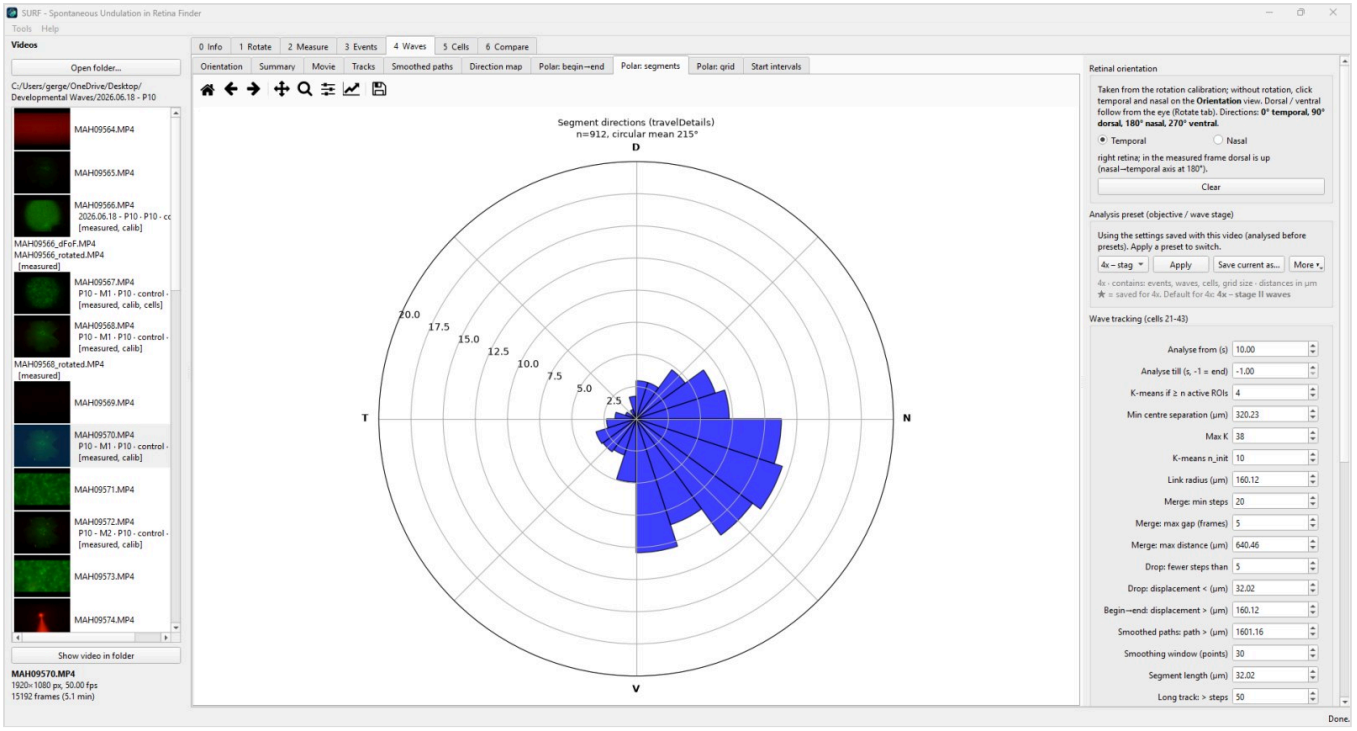
Plot	Shows
Tracks	every track, green start, red end
Smoothed paths	smoothed long tracks on the retina, with the nasal–temporal (purple) and dorsal–ventral (orange) axes
Direction map	mean local direction of travel in each region (red arrows) and wave start points (blue)
Polar: begin→end / segments / grid	direction distributions: per track, per path segment, per map region
Start intervals	time between wave starts



Smoothed paths: green = start, red = end; T–N and D–V axes drawn

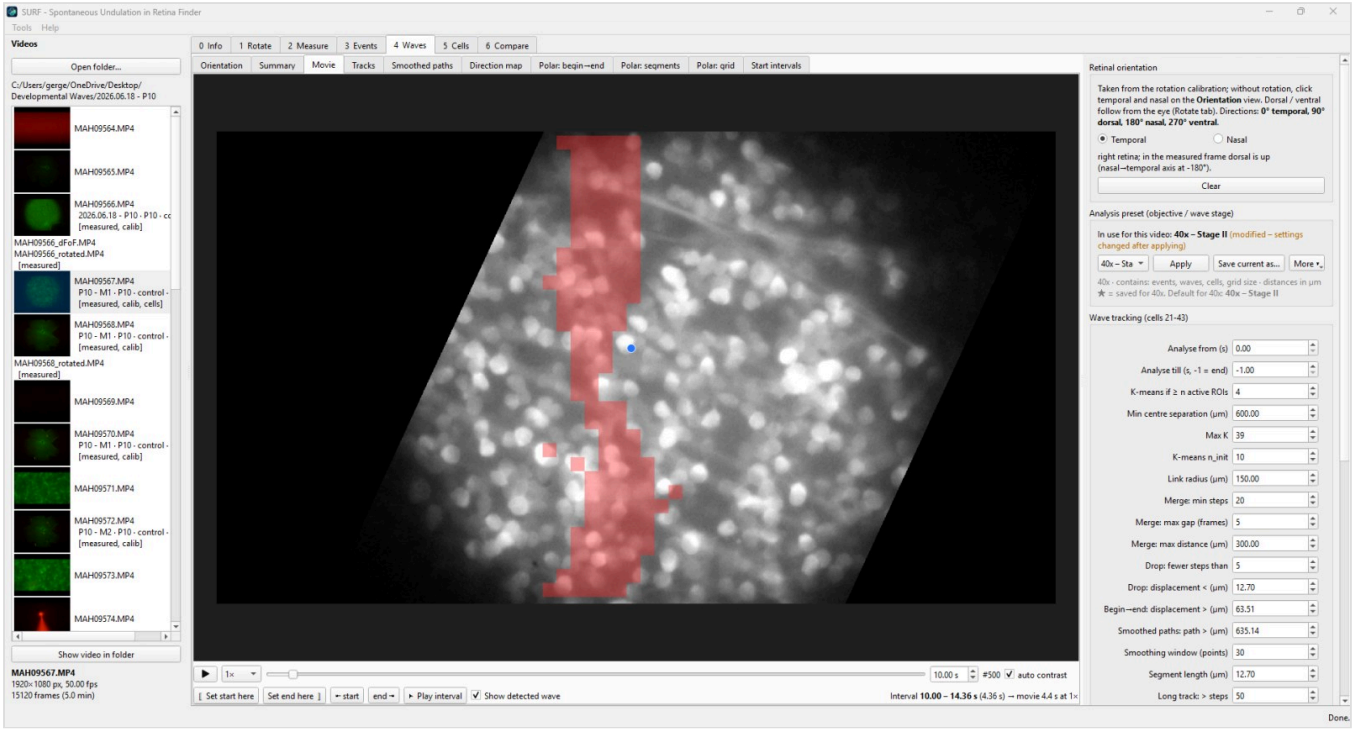


Direction map: local mean propagation direction and wave starts

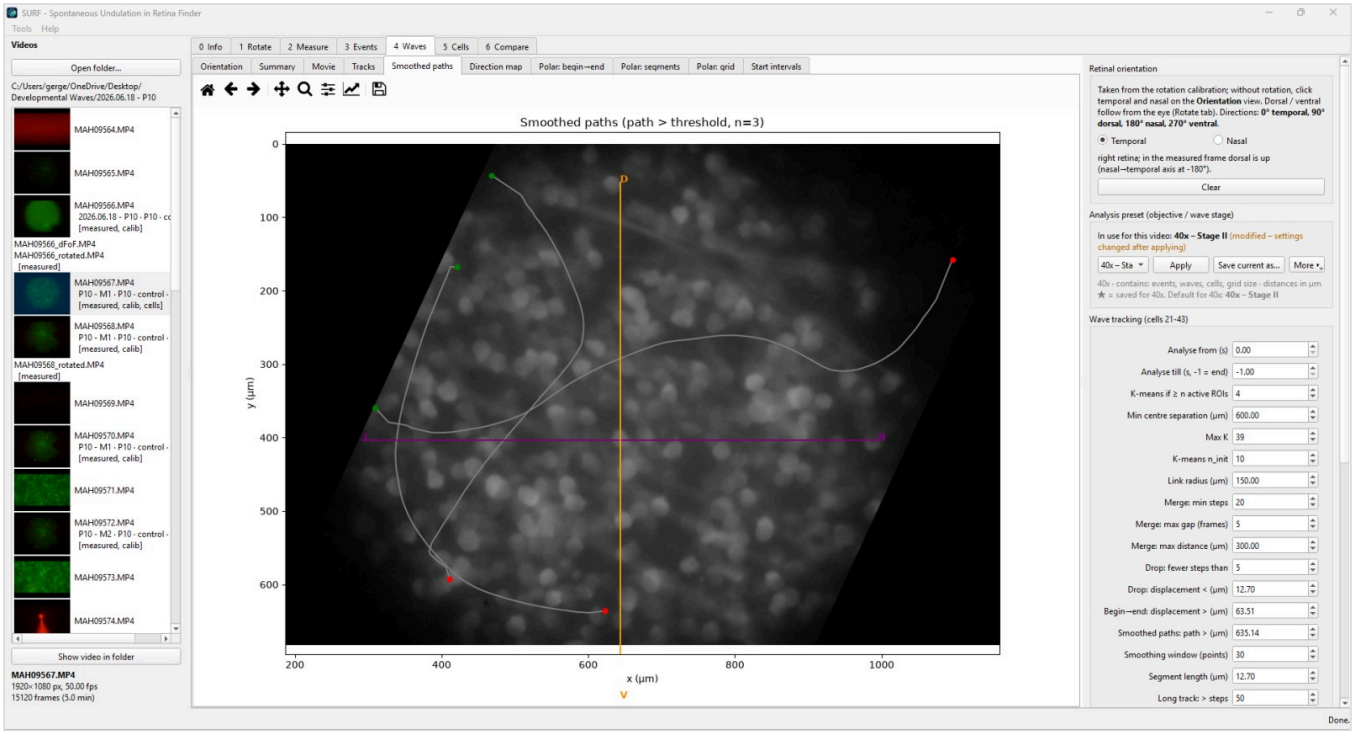


Polar plot of segment directions (T, D, N, V); here mostly towards nasal-ventral, mean 215°

At 40× one wavefront often fills the whole field as a long band. To get one centre for it instead of several, set **Min centre separation** larger than the band (here 600 µm), **Link radius** about 150 µm and **Merge: max distance** about 300 µm.



40× movie: one band-shaped wavefront with a single centre (Min centre separation 600 µm)



40× smoothed paths: three long tracks crossing the field

The mean-direction arrow is drawn only when the Rayleigh test says the directions are not uniform ($p < 0.05$).

Save results writes the `…_wave_results.xlsx` workbook and every plot as PNG, SVG, EPS and PDF. The figures are made for publication: Arial text stays editable, and every figure uses the same sizes (6 pt ticks, 7 pt labels). Place them at 100 % in Illustrator.

8. Tab 5 – Cells (40×)

The Cells tab finds single cells in the measured field and tests which of them take part in the waves found on the grid. Measure the grid and run the wave analysis first; the cells use the same rotated frame.

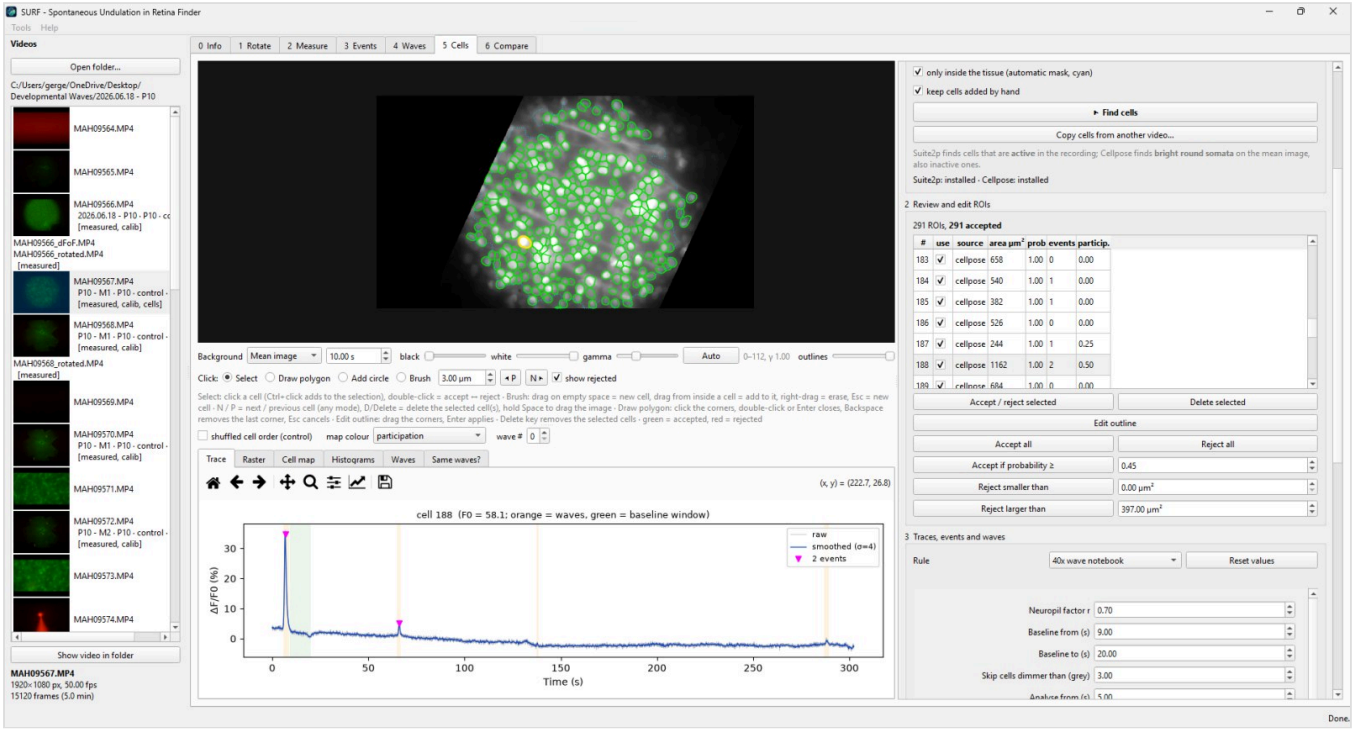
Step 1 – Find cells

Method	Finds	Notes
Suite2p (activity)	cells that are active during the recording	needs the add-on; *Suite2p threshold* lower = more cells
Cellpose (shape, mean image)	bright round somata, also inactive ones	needs the add-on; downloads its model on first use
FIJI RoiSet	ROIs drawn in ImageJ (.zip / .roi)	
By hand only	cells you draw yourself	

Options:

- **Use frames from / to**: the time range used for detection.
- **Leave out light-off frames**: frames with the light off are skipped.
- **Only inside the tissue** (cyan outline): no ROIs on the dark surround.
- **Keep cells added by hand**: a new detection does not remove your own cells.

Click **Find cells**. **Copy cells from another video…** copies the cells of another recording of the same field, and lines them up with this one automatically.



Cells tab after Cellpose: 291 cells (green), the selected cell in yellow, its ΔF/F trace below, settings on the right

Step 2 – Review and edit ROIs

Green outlines are accepted cells and red ones rejected. The table lists every cell with its source, area, probability, events and participation.

Click mode / key	Action
Select	click a cell; Ctrl+click adds to the selection; double-click switches accept ↔ reject
Draw polygon	click the corners; double-click or Enter closes; Backspace removes the last corner; Esc cancels
Add circle	a circle of the expected cell diameter
Brush	drag on empty space = new cell; drag starting inside a cell = add to it; right-drag = erase; set the brush size in µm
N / P	next / previous cell (the view follows)
Esc	let go of the current cell (next brush stroke = new cell)
D or Delete	delete the selected cell(s)
Space + drag	move the image
Edit outline / Apply outline	drag the corners of the selected cell's outline

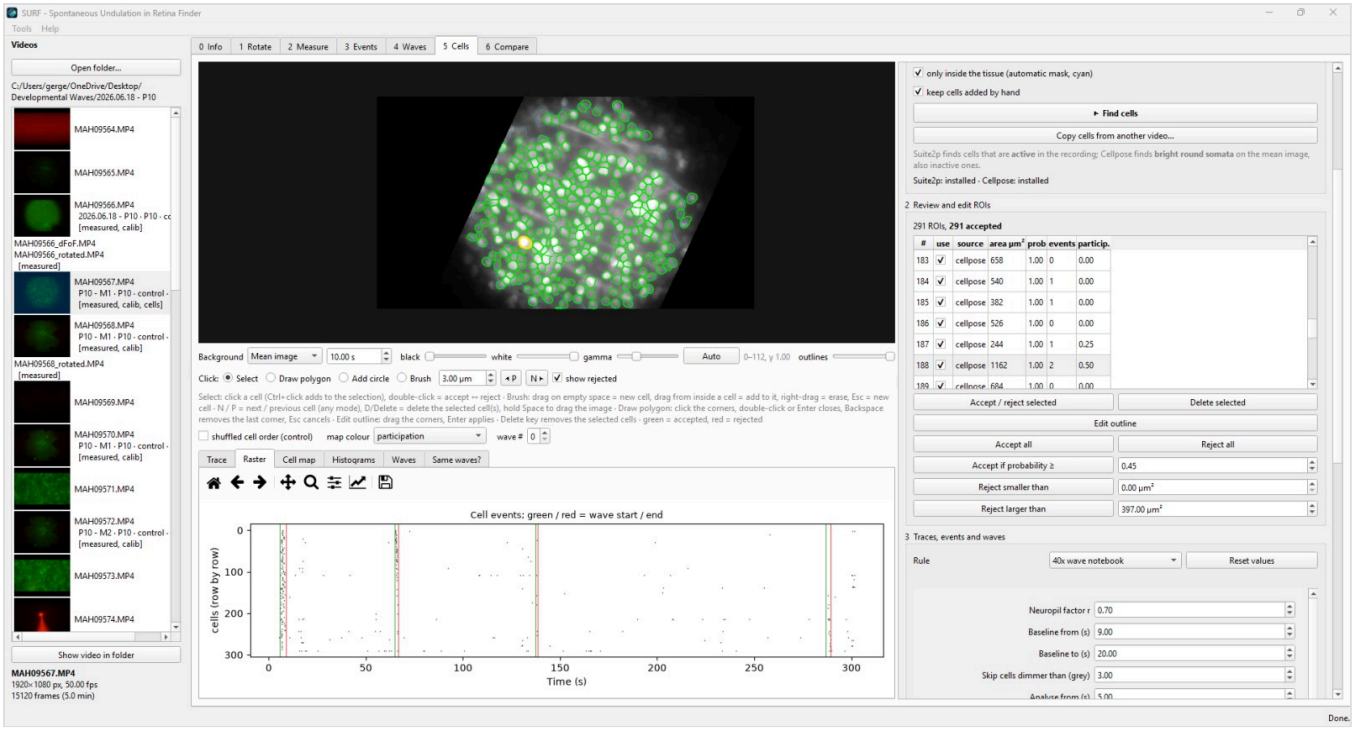
The buttons below the table accept or reject by probability, by size (smaller or larger than), or all at once. **Background** switches between mean, max and std images and video frames. **Black**, **white** and **gamma** set the brightness, and **outlines** sets how opaque the outlines are.

Step 3 – Traces, events and waves

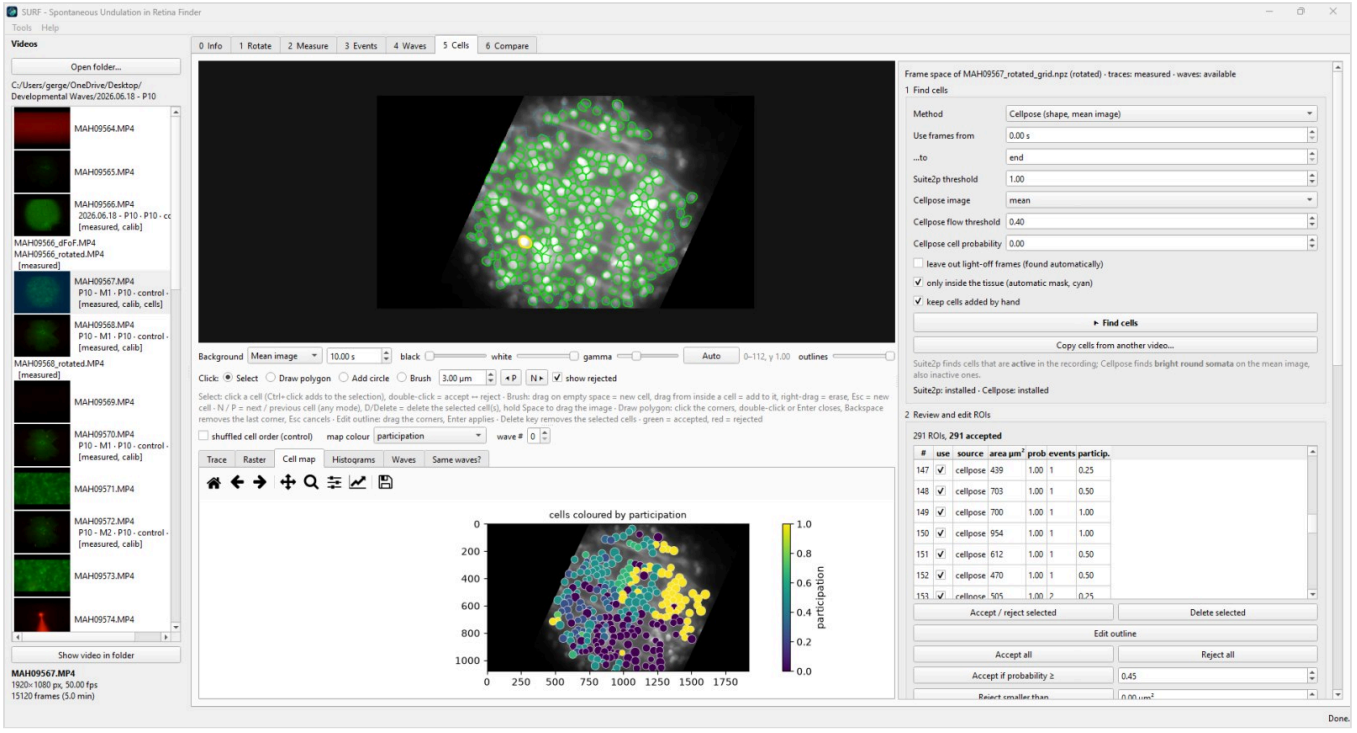
1. Click **Measure cell traces**. The neuropil ring around each cell is subtracted.
2. Choose a **Rule** for the events:
    - *Grid rule (tab 3 settings)*: the same peak rule as the grid. The settings start from the Events tab and can be changed here; changed values turn orange.
    - *One-cell notebook* or *40× wave notebook*: the rules of the original single-cell notebooks.
3. **Live preview on the selected cell** shows the effect of a setting immediately.
4. **Waves = 0 long tracks / 1 all tracks** sets which grid tracks count as waves for the cells. *1* is often better at 40×, where many waves cross the field quickly.
5. Click **Detect events + link to waves**.

The result line gives the number of cells, event rate, participation against chance (with p), the fraction of events inside waves, and whether the same cells join the same waves.

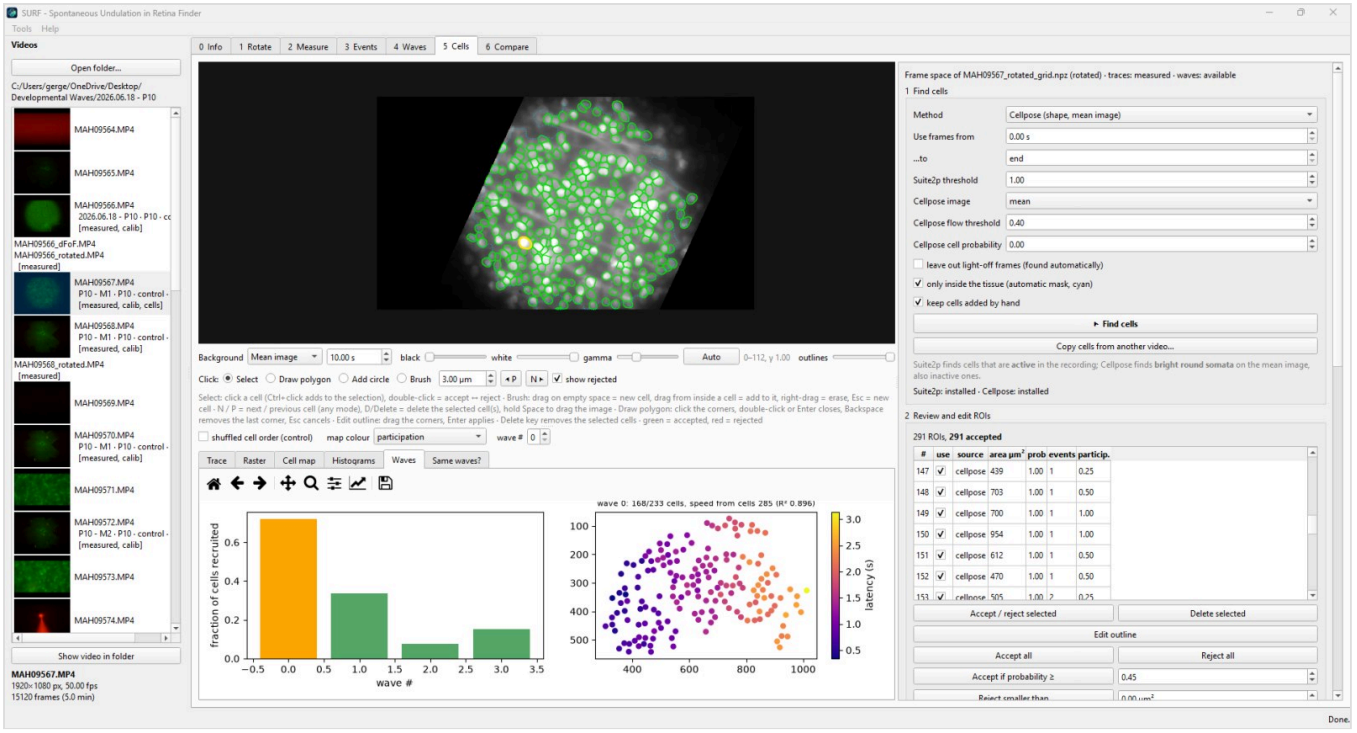
Plot	Shows
Trace	ΔF/F of the selected cell, its events, waves (orange bands), light-off periods (grey)
Raster	events of all cells; *shuffled cell order* is the control
Cell map	cells coloured by participation, rate or other measures
Histograms	distributions of rate, amplitude, decay and participation
Waves	recruitment per wave, and the latency map of one wave
Same waves?	cell pairs that join the same waves more than chance, compared with what their distance explains



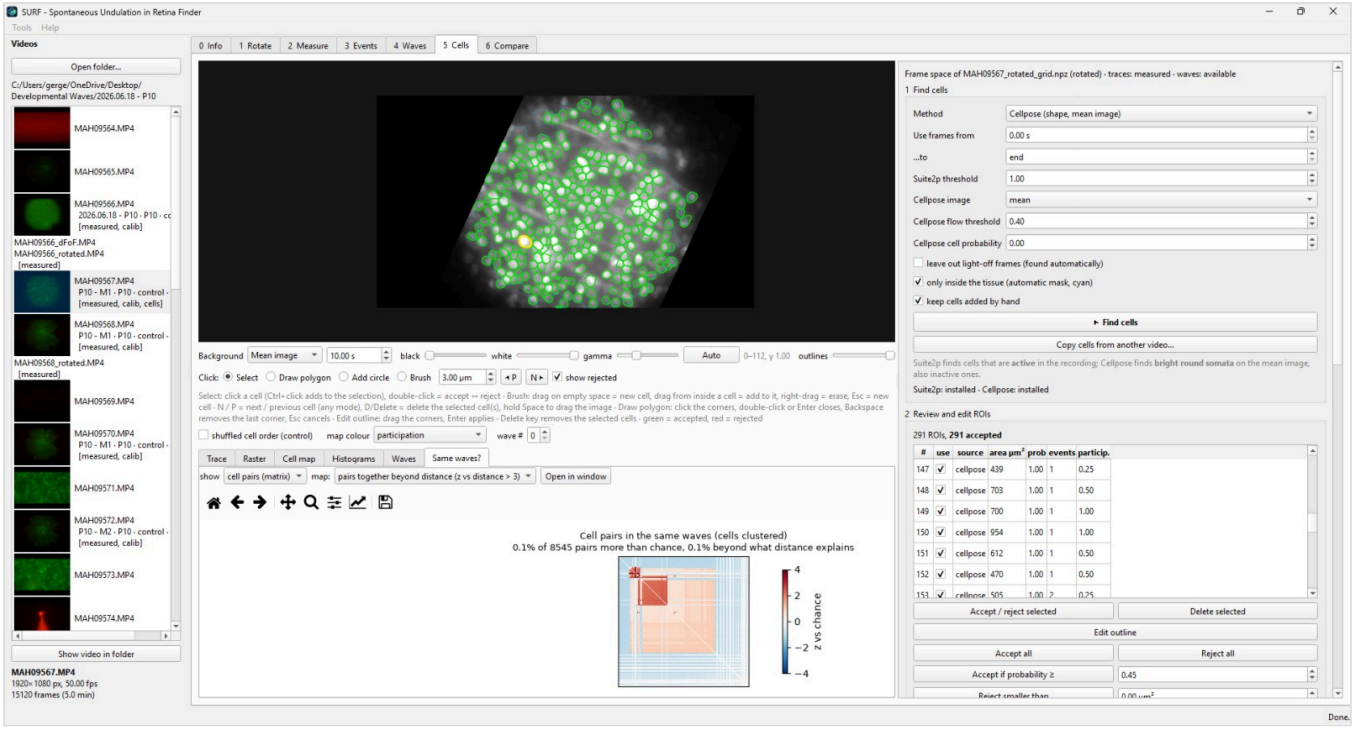
Raster of all cells: green and red lines mark the start and end of each wave



Cell map coloured by participation: cells in the upper right joined most waves



Waves plot: fraction of cells recruited per wave (left), and the latency of each cell in the selected wave (right), here 168 of 233 cells, speed from cells 285 µm/s



Same waves?: cell × cell matrix of pairs that join the same waves more than chance

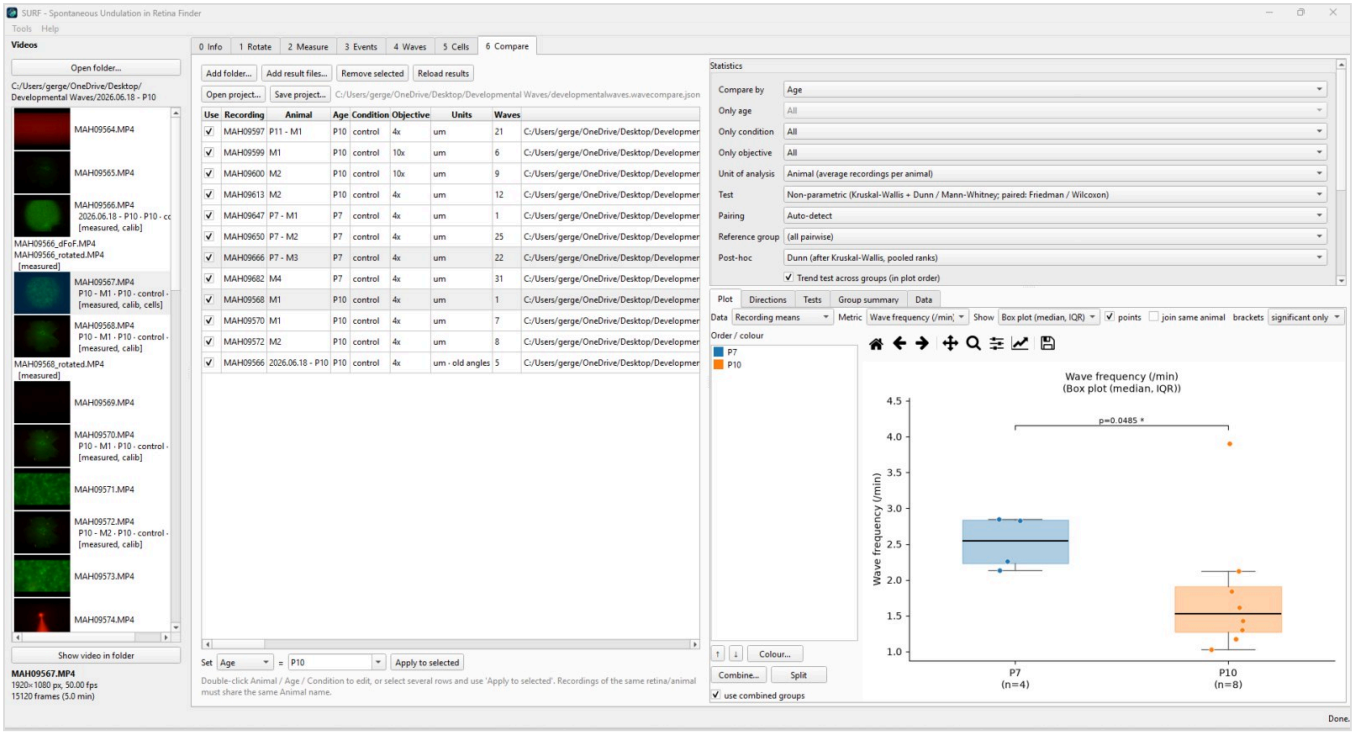
In **Same waves?**, **show** picks one panel at full size: cell pairs, waves × waves, distance, or map of pairs. **Open in window** shows all four large. Neighbouring cells naturally share waves, so *beyond distance* is the number to use when asking whether there are groups of cells that fire together.

Step 4 – Movie of cells and waves

Render cell movie… writes an MP4 in which each cell lights up at its events: **orange** for events in a wave, **cyan** for events outside waves, or coloured by ΔF/F. Active grid squares, wave centres and tracks can be added. *Follow brightness changes* keeps the image readable when the illumination changes.

9. Tab 6 – Compare

Compare pools the results of many videos and tests the differences between groups, for example ages or conditions. It works on the `…_wave_results.xlsx` workbooks, so run and save the wave analysis of every video first.



Compare tab: recordings on the left, statistics settings top right, plot of wave frequency P7 vs P10 below

Choosing recordings

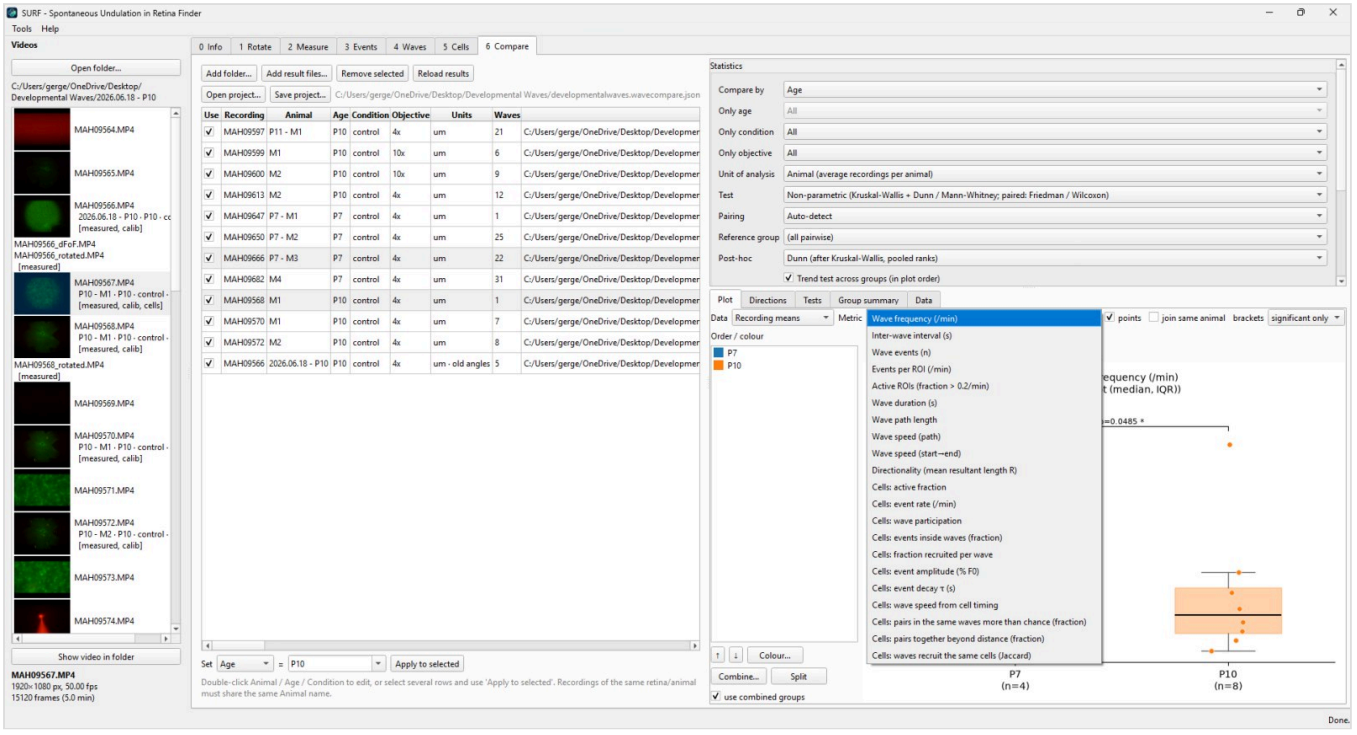
- **Add folder…** adds every analysed recording in a folder and its subfolders. **Add result files…** adds single workbooks.
- Only one result per video is used. If a video was measured with and without rotation, SURF keeps the one that matches its current setting.
- **Remove selected** and the **Use** tick boxes leave recordings out. **Reload results** reads the workbooks again after you re-analyse a video.
- **Save project… / Open project…** store the list and plot settings in a `.wavecompare.json` file.
- **Animal**, **Age** and **Condition** come from the Info tab. Double-click to edit them here; the change is written back to the video. For several rows at once, select them and use **Set … = … Apply to selected**.
- Recordings of the same animal must share the same Animal name, because the statistics treat the animal as one unit.

Statistics settings

Setting	Choices
Compare by	Age, Condition or another field
Only age / condition / objective	filter, e.g. only 4× recordings
Unit of analysis	**Animal** (recordings averaged per animal, recommended) or recording
Test	non-parametric (Kruskal–Wallis + Dunn / Mann–Whitney; paired: Friedman / Wilcoxon), parametric, or mixed model
Pairing	auto-detect when the same animals appear in several groups
Reference group	all pairwise, or each group against one control
Post-hoc	Dunn (pooled ranks) or Mann–Whitney
Trend test across groups	Jonckheere–Terpstra test for ordered groups such as ages, in plot order

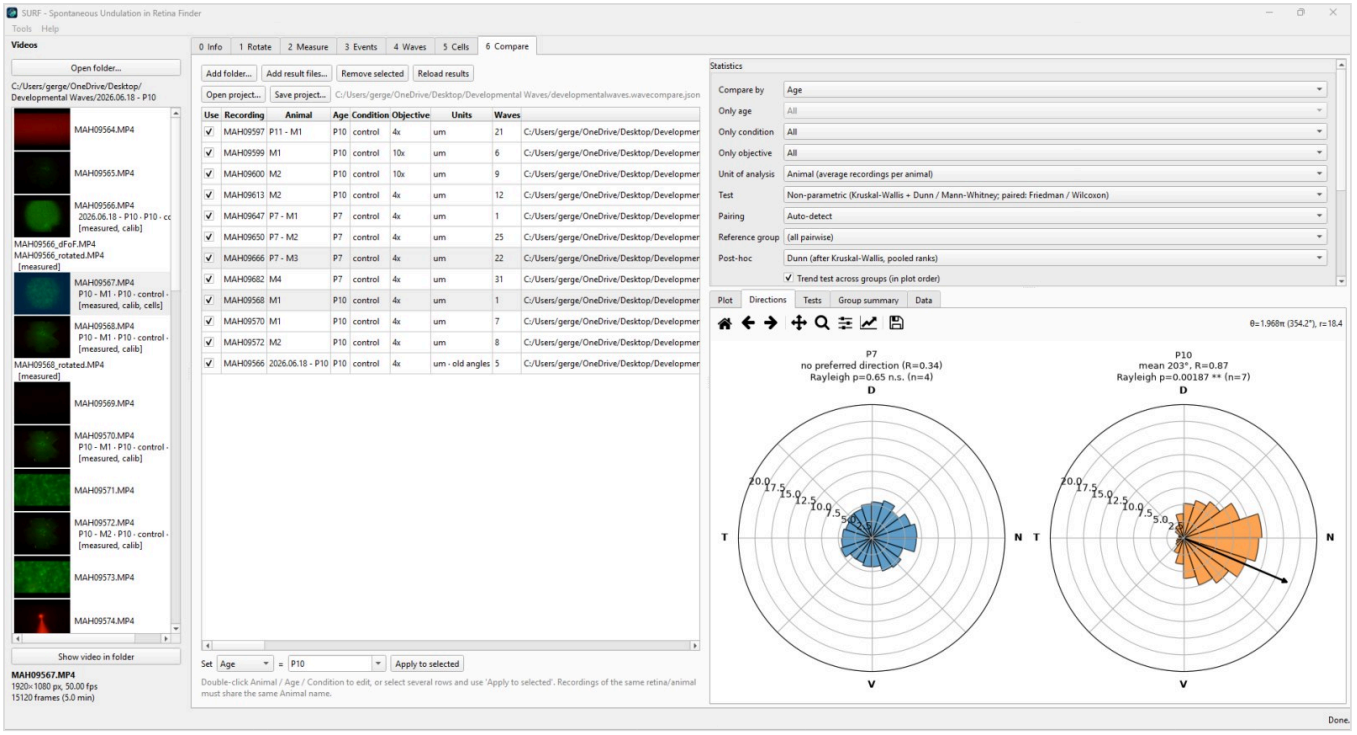
Plots

- **Data:** recording means, individual waves (violin plots), or individual cells.
- **Metric:** any wave or cell measure (list below).
- **Show:** box plot, bar plot or similar; **points** shows each animal or recording; **join same animal** connects paired values.
- **Brackets:** significant only (default), all, or none.
- **Order / colour:** move groups with ↑ ↓ and change their **Colour…**. **Combine…** pools groups, for example P7–P9 as *stage II*; **Split** separates them again; *use combined groups* switches between the two.

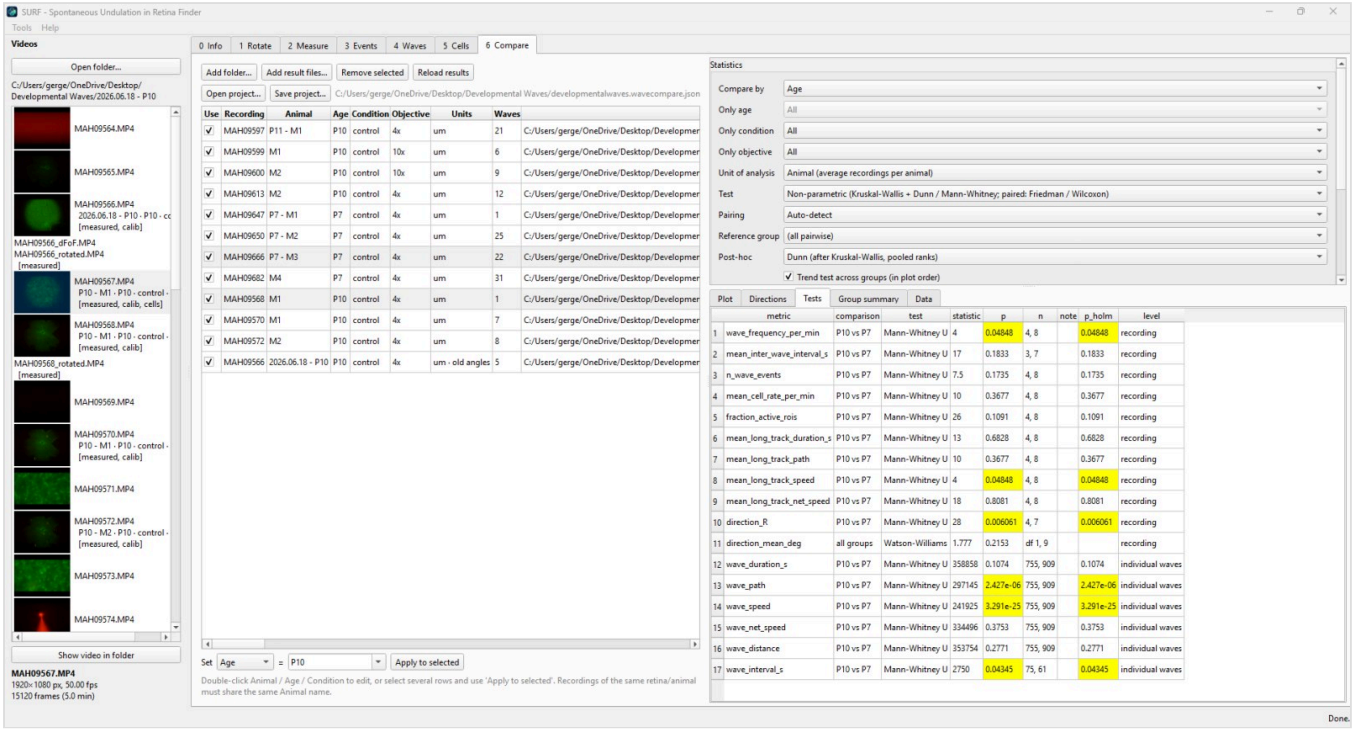


Metric menu: wave frequency, interval, duration, path, speed, directionality, and the cell measures

Directions, tests and export



Directions: polar plot per group, with the Rayleigh test; the arrow is drawn only when there is a preferred direction



Tests: every comparison with test, statistic, p, n and Holm-corrected p; significant values highlighted

- **Directions:** each group's distribution of wave directions, its mean direction, R and the Rayleigh p. Groups are compared with the Watson–Williams test.
- **Tests:** one row per metric and comparison. **level** says whether the test used recording/animal means or individual waves. Individual-wave tests have far larger n and should be read with care, because waves of one recording are not independent.
- **Group summary:** n, mean, SEM, median and IQR per group. **Data:** the values that went into the tests.
- **Export** writes an Excel workbook with all tables, and every plot as publication figures (PNG, SVG, EPS, PDF).

10. Tips and troubleshooting

Problem	What to do
One wave is split into several centres or tracks	raise **Min centre separation** (Waves tab), then **Link radius** and **Merge: max distance**
Separate waves are merged into one	lower **Min centre separation**
Too many events, noise counted	raise **Peak prominence** or **Smoothing σ** (Events tab); check with the live preview
Real events missed	lower **Peak prominence**; check **Ignore first (s)**
Flat trace with grey bands	SURF found light-off or dim frames. If that part is real data, set **Ignore light-off frames** to 0 (Events) or untick *leave out light-off frames* (Cells).
Distances shown in px instead of µm	choose the **Objective** and **Turret** on the Info tab, or calibrate the objective (Tools → Calibration tool)
Directions look mirrored or rotated	check the **Eye** and the temporal/nasal marks on the Rotate tab, then run the waves again
Compare shows two results for one video	press **Reload results**; one result per video is kept
Cell events are not linked to waves	run the wave analysis first; at 40× set **Waves = 1 all tracks**
macOS says the installer cannot be opened	System Settings → Privacy & Security → **Open Anyway**
An error message appears	click **Show Details…** and save the text; it names the line that failed

Good practice:

- Use one preset per objective and wave stage, and save it once it works. The preset name is written into every result.
- Check each analysis by eye in the **Movie** view before comparing groups.
- Give every video of one animal the same **Animal ID**.